



Watts and Wealth: Forecasting the Economic Pulse of Europe Through Electricity Consumption

Robin Kunju Mol Raj¹ · Marek Vochozka¹ · Yelyzaveta Apanovych¹

Accepted: 18 July 2025 / Published online: 11 September 2025
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Abstract

The association between electricity consumption and economic growth has evolved substantially in high-income countries, where energy-efficient transitions and sectoral shifts have contributed to a decoupling of the two. This study explores the predictive capacity of electricity consumption in forecasting GDP across six major European economies, Germany, France, Italy, Spain, the Netherlands, and Switzerland, using both traditional econometric techniques and deep learning models. While Granger causality tests, vector autoregression (VAR), impulse response functions (IRF), and forecast error variance decomposition (FEVD) reveal weak or negligible structural links between electricity and GDP, these findings support growing evidence of decoupling in advanced economies. In response, we implement Long Short-Term Memory (LSTM) neural networks to discover non-linear, temporal patterns in electricity data that can improve short-term GDP forecasting. To ensure robustness, we applied a walk-forward validation strategy using a 20-quarter rolling window. While in-sample performance appeared strong, out-of-sample forecasting revealed poor generalizability in most countries, highlighting the limitations of electricity data alone. Our findings underscore the limitations of linear models in decoupled economic systems and emphasize the role of electricity consumption as a potentially useful, but insufficient, predictive input when processed through non-linear machine learning frameworks. These results carry important implications for policymakers, energy analysts, and economists concerned with real-time economic monitoring and sustainability transitions in Europe.

Keywords Electricity consumption · GDP · LSTM · VAR · Europe · Forecasting

JEL Codes C53 · Q47 · E37

1 Introduction

Understanding the complex relationship between electricity consumption and economic growth (Yang, 2010) is crucial for designing effective and sustainable energy policies, particularly within Europe's leading economies (Sadorsky, 2012). As a core input to productivity, infrastructure, and quality of life, electricity consumption (Cui et al., 2022), whether generated domestically or imported, serves as a fundamental driver of national development. Investigating the direct linkage between electricity usage and economic performance is especially pertinent in highly industrialized nations, where energy systems and economic growth are tightly coupled. Numerous studies have examined this nexus using a range of methodological approaches across diverse economic contexts (Abbasi et al., 2021; Arcabi et al., 2021; Dey & Tareque, 2020). For example, residential electricity consumption in Cameroon has been linked to causal dynamics and sector-specific policy recommendations for sustainable development (Guefano et al., 2023). In Colombia, electricity consumption inequality has been analyzed through per capita metrics and Gini coefficients across different socioeconomic and geographic groups (Cabello Eras et al., 2022). From a broader perspective, studies in Peru and Egypt confirm long-run causality between electricity consumption and GDP, with short-run influences from demographic and inflationary factors (Navarro et al., 2023; Han & Radwan, 2021).

High-frequency data analyses, such as in Bangladesh, have disclosed strong correlations between electricity usage and macroeconomic activity at national and subnational levels (Arshad & Beyer, 2023). In Pakistan, sectoral electricity consumption has been linked to energy intensity and economic decoupling trends over time (Lin & Raza, 2021). In the OECD, elasticity estimates show a declining relationship between electricity demand and GDP, attributed to electrification saturation (Liddle et al., 2023), while case studies such as the 2021 Texas power crisis highlight the economic vulnerability linked to energy system resilience (Liu et al., 2024). Further, country-specific research in Iran using Structural Vector Autoregression (SVAR), Directed Acyclic Graphs (DAGs), and Time-Varying Parameter Vector Autoregression (TVP-VAR) models uncovers the impact of structural macroeconomic shocks on electricity consumption (Esmacili & Rafei, 2021). Wavelet and fuzzy regression analysis also suggest that the energy-GDP nexus is highly sensitive to both timescales and development levels across different nations (Pandelara et al., 2022). In the same way as lower economy countries, the studies were done in higher economy countries like China (Wu et al., 2021; Zhang et al., 2020). In the age of digitization and energy transformation, knowing the relationship between electricity and GDP is essential because it facilitates real-time economic monitoring, informs sustainability policies, and directs effective energy planning, particularly amid post-COVID recovery and market volatility.

Current research on the electricity-GDP nexus faces several challenges, including weakened causality in high-income economies, a narrow forecasting focus, insufficient handling of nonlinear and seasonal dynamics, and methodological difficulties in aligning high-frequency electricity data with low-frequency GDP indicators. Building on this global literature, the present study looks into the relationship between electricity consumption and GDP in the context of the top six European economies. It

applies both traditional econometric models and advanced machine learning methods to better understand and predict this relationship, leading to the first research question:

RQ1: To what extent does electricity consumption predict GDP growth in major European economies, and how effectively can machine learning models capture this relationship compared to traditional econometric approaches?

Given the increasing complexity and interdependence of energy and economic systems (Benkraiem et al., 2019; Caraianni et al., 2014; Saldivia et al., 2020), it becomes essential to assess the comparative predictive performance of these modeling approaches, prompting the second research question:

RQ2: How does the predictive performance of machine learning models, specifically LSTM neural networks, differ from traditional econometric models in capturing the dynamics between electricity consumption and GDP in top European economies?

This study addresses a critical gap by exploring electricity consumption's predictive value in economies where causality with GDP has weakened. Unlike prior studies focused solely on structural analysis, we combine econometric diagnostics with LSTM to improve short-term GDP forecasting accuracy. The novelty lies in positioning electricity usage as a practical, non-causal input for nowcasting GDP in high-income European countries. The selection of these 6 European countries, including Germany, France, Italy, Spain, the Netherlands, and Switzerland, is because of all high-income European economies with reliable electricity consumption and GDP data. Their changes in industrial composition, energy dependence, and degrees of EU electricity market integration offer a robust basis for comparative analysis.

The relationship between electricity consumption and economic growth has been widely studied across nations and methodological frameworks, but the results remain context-dependent and diverse. Early work Kraft and Kraft (1978) sparked an ongoing debate on whether energy consumption drives economic performance (Georgescu et al., 2024; Oprea et al., 2025) or vice versa. Several studies have confirmed the premise that electricity consumption drives GDP, as evidenced in Japan (Ikegami & Wang, 2016), Burkina Faso (Ouedraogo, 2010), and newly industrialized countries (Azam et al., 2021). In contrast, countries like Germany show a reverse causal direction, where GDP influences electricity consumption (Ikegami & Wang, 2016), while Malaysia (Chandran et al., 2010), Egypt (Han & Radwan, 2021), and Peru (Navarro et al., 2023) show unidirectional or conservation-based dynamics, with real GDP causing power demand. Bidirectional causality or feedback effects dominate the literature, specifically in OECD countries (Ohler & Feters, 2014), global panels (Narayan et al., 2010), U.S. states (Saldivia et al., 2020), and African nations (Ouedraogo, 2013), substantiating the idea of mutual interdependence between power use and growth. Spatial and sectoral heterogeneity complicates this linkage; for example, China's electricity and GDP relationship reveals region-specific decoupling patterns (Hao et al., 2019) and spatial dependency formed by industry structure, FDI, and R&D (Cui et al., 2021).

From a methodological standpoint, diverse approaches have been accustomed to uncovering these dynamics. Techniques such as the Autoregressive Distributed Lag (ARDL) model, Vector Error Correction Model (VECM), cointegration tests, wavelet analysis, and Generalized Method of Moments (GMM) account for structural breaks, temporal variations, and heterogeneity. For example, the Fourier-based ADL cointegration test provides robust evidence of long-run relationships in EU countries (Arcabi et al., 2021), while wavelet-based models disclose time-varying fuzzy causality across 61 countries (Pandelara et al., 2022). Similarly, threshold models demonstrate that the direction of causality in OECD countries changes with income levels (Tran et al., 2022). Studies using panel ARDL frameworks highlight the long-run impact of renewable energy on growth and environmental outcomes in the Mediterranean (Belaid & Zrelli, 2019) and industrialized economies (Azam et al., 2021), reinforcing the sustainability narrative. Recent crises like COVID-19 and the Ukraine war increased electricity price volatility (Bâra et al., 2024), reinforcing the importance of robust machine learning forecasts.

The role of energy composition is also fundamental. Research in the U.S. and China finds fossil and renewable energy sources affect GDP differently, with sector-specific causal directions (Korkmaz, 2022; Rahman et al., 2020). Mexico is a rare case where there is no causality, emphasizing the importance of an infrastructure-driven, self-sufficient strategy (Massa & Rosellon, 2020). Meanwhile, nations such as Brazil (Hdom & Fuinhas, 2020) and Cameroon (Guefano et al., 2023) prioritize residential and sectoral variables, with energy supply, price, and urbanization all influencing economic and environmental consequences. The IKoMet model (Kolin et al., 2021) adds a predictive dimension to the debate, offering a tool for long-term forecasting of electricity-GDP trends. The role of artificial neural networks (Colak & Koy, 2023; Koy & Çolak, 2023) play a major role in prediction.

In sum, the literature reflects a multifaceted and regionally diverse energy growth nexus, with strong evidence supporting context-sensitive policy frameworks. Whether driven by causality direction, energy type, or socio-economic structure, the relationship remains essential for sustainable development, especially in energy-dependent and rapidly growing economies like Bangladesh (Zhang et al., 2021), India (Ghosh, 2009; Wu et al., 2019), and Saudi Arabia (Alsaedi & Tularam, 2020). While previous studies have examined the electricity-GDP nexus using a variety of econometric tools, little research has compared them to machine learning approaches. To close this gap, the current study combines classic econometric methodologies with LSTM neural networks, allowing for causal analysis and prediction evaluation across six major European economies.

The rest of this paper is organized as follows: Sect. 2 details the data sources, aggregation methods, and the dual-path approach that combines both traditional econometric techniques and LSTM neural networks. Section 3 discusses the empirical results, including stationarity tests, causality analysis, VAR diagnostics, and LSTM forecasting performance. Section 4 offers a comparative analysis, economic implications, and their relevance to forecasting and sustainability policy. Finally, Sect. 5 summarizes the main findings and offers suggestions for future research.

2 Methodology

2.1 Data

For this analysis, Electricity consumption is hourly, and the GDP will be monitored quarterly; these data (Kunju Mol Raj, 2025) were obtained from the period 2015 to 2025 and from ENTSO-e (European Network of Transmission System Operators for Electricity) and Eurostat Websites, respectively. The electricity consumption values are labeled as MW (Megawatts), and the GDP is in billion Euros. The six nations were chosen based on their nominal GDP rankings in the fourth quarter of 2024, as determined by Eurostat statistics. This guarantees that the research is limited to Europe's most economically significant nations as of the most recent available data point, covering the most relevant patterns between electricity use and economic activity. Germany, France, Italy, Spain, the Netherlands, and Switzerland are among Europe's top economies.

2.2 Methods

In our empirical investigation, two complementary analytical methodologies are used. Because of its wide econometric modeling capabilities, VAR (Koy & Çolak, 2024) is implemented in Python, employing well-established modules such as statsmodels. Concurrently, the LSTM neural network analysis is carried out using Wolfram Mathematica, which has robust, symbolic, and neural-net-specific functions. This dual-tool strategy allows us to efficiently use the characteristics of each software platform while also ensuring strong and detailed analytical validation. Johansen and ARDL models are widely used for long-run analysis; however, their application was constrained here due to data frequency mismatch and limited quarterly observations. Given these limitations, we focused on short-run dynamics via VAR and on predictive modeling through LSTM, which better accommodates nonlinear patterns in high-frequency datasets. The econometric analyses were conducted using Python due to its flexibility and rich statistical libraries, while the LSTM neural network models were implemented in Wolfram Mathematica. This dual-platform approach was chosen for Mathematica's symbolic computation capabilities and its native support for time series neural network structures. The overall analytical framework follows two independent paths, econometric analysis and machine learning prediction, applied sequentially rather than in combination. This structure is illustrated in Fig. 1

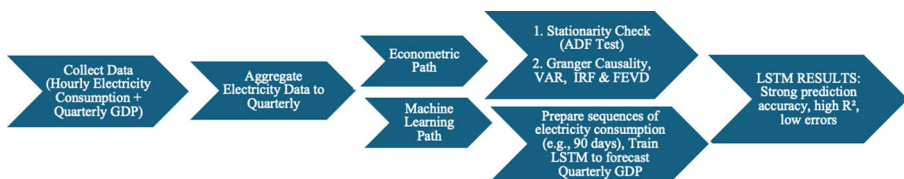


Fig. 1 Methodological flowchart illustrating sequential use of econometric and machine learning approaches

At first, the relationship between electricity consumption and GDP can be found using the traditional statistical model, the Pearson correlation coefficient, for that the electricity consumption data is aggregated into quarterly with the GDP.

Next, The Augmented Dickey-Fuller (ADF) test was applied to both electricity consumption and GDP time series to examine the presence of unit roots. The test was specified as:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \epsilon_t \quad (1)$$

where y_t represents either electricity consumption or GDP. α , βt , γ , $\sum_{i=1}^p \delta_i \Delta y_{t-i}$, and ϵ_t are the constant (drift) term-accounts for non-zero mean, time trend (optional) – included if there's a deterministic trend, key coefficient-tests for unit root, lagged differences to account for autocorrelation, and white noise error term respectively. The null hypothesis of a unit root was tested against the alternative of stationarity. Lag length p was selected based on the Akaike Information Criterion (AIC).

The Granger causality test was used to determine whether electricity consumption can predict GDP and whether GDP can predict electricity consumption over time. This test helps identify the direction of causality between the two variables using their past values. The basic model is written as:

$$Y_t = c + \sum_{i=1}^p \alpha_i Y_{t-i} + \sum_{j=1}^q \beta_j X_{t-j} + \epsilon_j. \quad (2)$$

Where, Y_t is the dependent variable (Predicted variable, here, GDP or Electricity consumption data), X_t is the independent variable (the predictor, Electricity consumption or GDP data), c is the constant, α_i and β_j are the coefficients, p and q are the number lags for Y and X , respectively, and ϵ_t is the error term. And the Hypothesis is written as, Null Hypothesis (H_0) is the X does not Granger-cause Y (the coefficients $\beta_j = 0$ for all j) and the Alternative Hypothesis (H_a) is the X Granger-causes Y (at least one $\beta_j \neq 0$). The test statistic is typically calculated using an F-test:

$$F = \frac{(RSS_R - RSS_{UR})/q}{RSS_{UR}/(n - p - q - 1)}. \quad (3)$$

Where, RSS_R is the residual sum of squares for the restricted model, RSS_{UR} is the residual sum of squares for the unrestricted model, n is the number of observations, p is the number of lags used in the unrestricted model, q is the number of lags for X . And the calculated F-statistic is then compared against a critical value from the F-distribution to determine whether to reject the null hypothesis.

Given the focus on short-run dynamics and the stationarity confirmed via the ADF test, Johansen cointegration testing was not performed. Instead, the VAR model was applied to the differenced series based on the results of the Granger causality test. While Granger causality remains a widely used method for detecting directionality

in time series, its validity is limited in settings with nonlinear or regime-switching dynamics-conditions that are likely present in energy-economic systems of high-income countries. Recognizing these limitations, we incorporated LSTM neural networks to better capture complex, nonlinear dependencies without relying on strict linear assumptions.

VAR Model The VAR model is used to study the short-term dynamic connection between electricity consumption use and GDP. This model considers both variables to be endogenous and represents their interdependence using their respective lagged values. Prior to estimate, the ADF test is accustomed to check the variables' stationarity after initial differencing. Standard information metrics, such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), are used to determine the best lag duration. Once the lag structure has been defined, the VAR model is estimated on the differenced series to prevent false regression caused by non-stationarity. Model diagnostics, such as residual autocorrelation tests and stability checks (using the companion matrix's characteristic roots), are used to guarantee the results' validity. The computed VAR model forms the basis for further Impulse Response Function (IRF) and FEVD investigations.

LSTM An artificial neural network (NN) approach called LSTM is used to forecast the future development of GDP. The parameters necessary for the creation of the Neural Network are listed below:

The number of elements in the Neural Network Model depends on the model's configuration and the input data (electricity consumption data, daily aggregated), with a matrix size of $m \times n$; in this case, it varies depending on the input data, i.e., " 90×1 ", " 90×2 ", " 90×3 ", etc. Diverse activation functions (utilizing the Elementwise Layer) are accustomed to facilitating signal propagation among NN layers. The selection of activation functions encompasses the following options: Sin, Ramp (also recognized as ReLu – Rectified Linear Unit), and ArcTan. Time series lag is a sequence of previous values used to predict future values. The lag is set to be 90 days. The NN structure is the same in all circumstances (except for the parameter changes). Figure 2 shows a schematic of the NN.

The neural network consists of six main layers in addition to the input and output.

The six layers between the input and output layers are hidden. In the Input Layer, A matrix representing electricity consumption data is provided in the input layer. Different sizes of matrices will be investigated in the experiment (based on the input data). Here, the matrix size is " 90×1 ". The value 'm' in the matrix represents the number of consecutive variables required to compute the following variable. Next, the first hidden layer is an LSTM layer. It generates an output matrix of size $m \times n$, where 'm' is the time series lag and 'n' is chosen empirically. Here, the matrix at the input layer 90×1 will generate a matrix 90×90 at the output). The 'n' value grows by one. Next, is the 2nd Hidden Layer (Elementwise Layer - Perceptron), which is a simple network that functions as a perceptron. These layers' activation functions will be chosen at random from a specified list of activation functions. 3rd Hidden Layer (Elementwise Layer), with the same activation function selection criteria as the

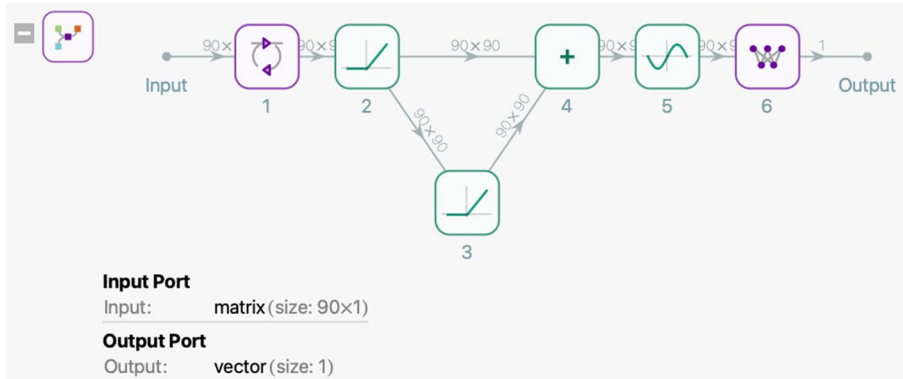


Fig. 2 Structure of NN with LSTM layer

second hidden layer. The 4th Hidden Layer (Plus Layer), known as the “Plus” layer, accomplishes summing. It receives input from the second and third hidden layers and transmits the output to the fifth hidden layer. 5th Hidden Layer (Elementwise Layer), uses the same activation function selection criteria as the second and third hidden layers. 6th Hidden Layer (Linear Layer), which acts on a data matrix at the input (a 90×90 matrix is shown in the figure). It generates a vector with one element as output. And the final output layer predicts the macroeconomic variable (GDP).

Long-Short Term Memory Layer [n] represents a network that receives an input matrix representing a sequence of vectors and outputs a sequence of the same length. Each input sequence element is a vector of size k, and each output sequence element is a vector of size n. The LSTM consists of four blocks: Input gate, output gate, forget gate, and memory gate (Krulický et al., 2024).

$\{x_1, x_2, \dots, x_T\}$ is the input sequence, and the output of the LSTM is a series of states $\{s_1, s_2, \dots, s_T\}$. The cell state is defined as follows:

$$c_t = f_t * c_{t-1} + i_t * m_t \quad (4)$$

where

c_t is a new variable state, f_t is the forget gate, c_{t-1} is the the initial state of the variable, i_t is the input gate, and m_i is the memory gate. The input gate is defined as follows:

$$i_t = \sigma [W_{ix}x_t + W_{is}S_{t-1} + b_i]. \quad (5)$$

Where, $\sigma(z) = 1/(1 + \exp(-z))$, σ is Logistic Sigmoid, W_{ix} is an input weight in the input gate matrix $n \times k$, x_t is an input variable, matrix $n \times k$, W_{is} Weight of the state in the input gate, matrix $n \times n$, S_{t-1} is the initial state, b_i is the bias, vector size n . The state is defined as follows:

$$s_t = o_t * \text{Tanh}[c_t] \quad (6)$$

where, s_t is a state of the variable, o_t is the output gate, and $Tanh$ is the Hyperbolic tangent. Output gate is defined as follows:

$$o_t = \sigma [W_{ox}x_t + W_{os}S_{t-1} + b_o] \quad (7)$$

where, W_{ox} defines the input weight in output gate, matrix $n \times k$, W_{os} Weight of the state in output gate, matrix $n \times n$, and b_o is the bias, vector size n . When compared to, e.g., a Gated Recurrent Layer, the benefit of LSTM is in the forget gate:

$$f_t = \sigma [W_{fx}x_t + W_{fs}S_{t-1} + b_f] \quad (8)$$

where, W_{fx} is an input weight in forget gate, matrix $n \times k$, W_{fs} is the weight of the state in forget gate, matrix $n \times n$, and b_f is the bias, vector size n . The main processes of LSTM include memory gate as follows:

$$m_t = Tanh[W_{mx}x_t + W_{ms}S_{t-1} + b_m] \quad (9)$$

where, W_{mx} Defines the input weight in memory gate, matrix $n \times k$, W_{ms} , Weight of the state in memory gate, matrix $n \times n$, and b_m is the bias, vector size n . The LSTM models were trained and evaluated on the entire dataset due to the limited number of quarterly GDP observations available for each country.

Walk-Forward Validation Strategy To mitigate potential overfitting and provide a reasonable assessment of predicting ability, we used a walk-forward validation technique. At each stage, the LSTM model was trained on a 20-quarter rolling window and assessed in the next quarter. Evaluation measures (RMSE, MAE, MAPE, R^2) were used to assess out-of-sample accuracy after reinitializing and retraining the model for each step.

3 Results

Daily power consumption statistics for all six countries (Germany, France, Italy, Spain, the Netherlands, and Switzerland) show strong seasonal trends and cyclical swings from 2014 to 2025. Each country's power consumption has an annual cycle, with distinct peaks and troughs indicating the influence of weather, industrial cycles, and potentially public holidays. The total amount of consumption varies greatly between nations due to differences in population, industrial structure, and energy efficiency. France and Italy have greater average usage than smaller economies like Switzerland. Across all nations, there is no clear visual evidence of long-term increasing or negative trends, reflecting relative stability in power consumption over time, however short-term volatility and seasonality remain significant. This behavior suggests that stationarity checks, differencing, or transformation should be performed prior to continuing with time series modeling techniques such as VAR or machine learning. In Fig. 3, it's clear that Germany and France show cyclical amplitude, showing the sensitivity of the industrial sector's electricity consumption, in the

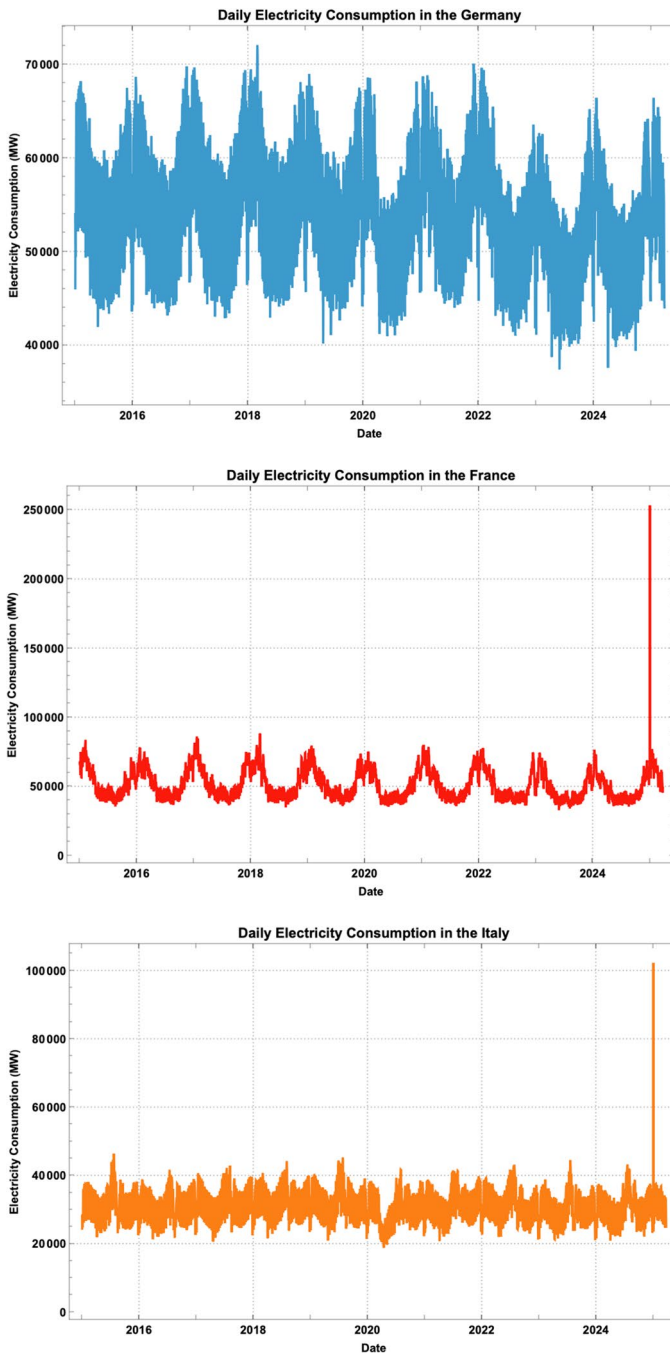


Fig. 3 Time series plot showing seasonal fluctuations in electricity consumption of the top 6 European economies. Note. This figure illustrates the seasonal variation in electricity consumption over time. The recurring peaks and troughs reflect regular patterns likely influenced by climatic or economic cycles

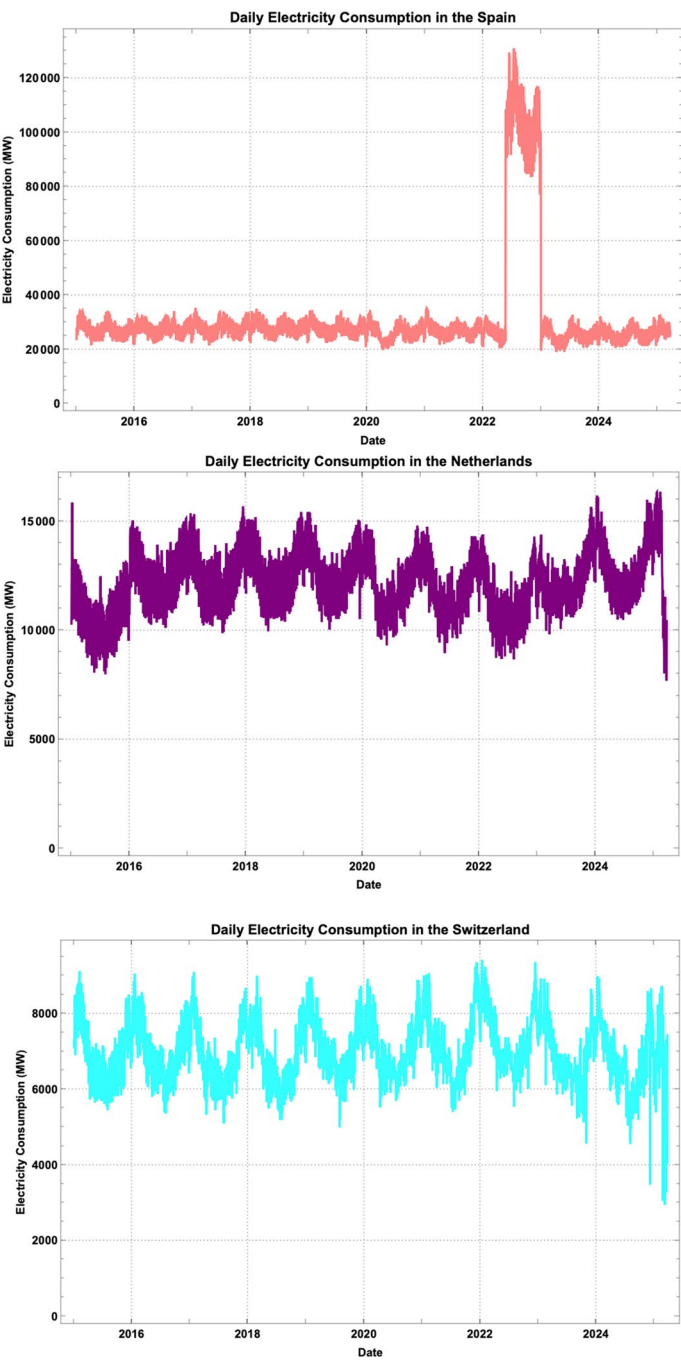


Fig. 3 (continued)

same way as when we look into the Netherlands and Switzerland, a stable usage of electricity consumption can be noticed. Italy, France, and Spain have a sudden fluctuation in the period between 2022 and 2025.

Germany, France, Italy, Spain, the Netherlands, and Switzerland's GDP trajectories show generally comparable increasing patterns from 2014 to 2025, indicating a consistent economic expansion across leading European nations. Notably, all nations had a steep recession in 2020, in line with the economic shock caused by the COVID-19 pandemic, followed by a robust and speedy rebound. Following the recovery, GDP growth rates continue to rise, demonstrating these economies' durability and flexibility. The growth looks to be smooth and consistent, indicating well-managed macroeconomic circumstances and structural stability. Switzerland is an outlier, with more erratic and step-by-step GDP growth, most likely due to variances in data granularity, measurement, or economic structure. The overall continuous upward trend of GDP in the other five nations allows the use of time-series models, allowing for comparative econometric research to investigate causal linkages with electricity consumption. These visual patterns provide a compelling platform for investigating energy-economy dynamics, notably the possible decoupling of economic development and energy consumption in mature nations (Fig. 4).

Correlation between Electricity Consumption and GDP The cross-country correlation across the top six European economies reveals a generally weak and more heterogeneous relationship between electricity use and GDP. In Germany, there is a moderate negative correlation ($r = -0.45$), implying a decoupling effect in which economic growth does not come with proportionate rises in electricity usage—presumably due to energy efficiency improvements and structural reallocation towards less energy-intensive industrial sectors. For France, a weak negative relationship ($r = -0.21$) applies, while the same applies to Italy and Switzerland, where associations are close to zero but marginally negative ($r = -0.08$). By contrast, Spain and the Netherlands display marginally positive associations ($r = 0.18$ and $r = 0.05$, respectively), implying a moderate co-movement between economic activity and energy demand. These varied results imply generally a more complex, country-specific modeling exercise in identifying possible underlying causality beyond simple linear associations (Fig. 5).

The ADF test outcomes reveal the GDP series at a level that is not stationary in all six countries, requiring transformation in time series modeling. The log difference in GDP is stationary in all the cases, showing suitability for VAR as well as Granger causality testing. The consumption of electricity behaves more heterogeneously: whereas it is stationary at a level in Italy and Spain, in the rest of the countries, it is not stationary. After the first difference in the log of the consumption of electricity, Germany, France, and Italy become stationary, while Spain, the Netherlands, and Switzerland become stationary after a second difference. The findings reflect structural and country-specific heterogeneities in the underlying electricity consumption dynamics, illustrating the role data preprocessing plays in ensuring sound econometric modeling (Table 1).

The Granger causality test among six European economies reveals no evidence of electricity consumption Granger-causing GDP growth in any country. This means

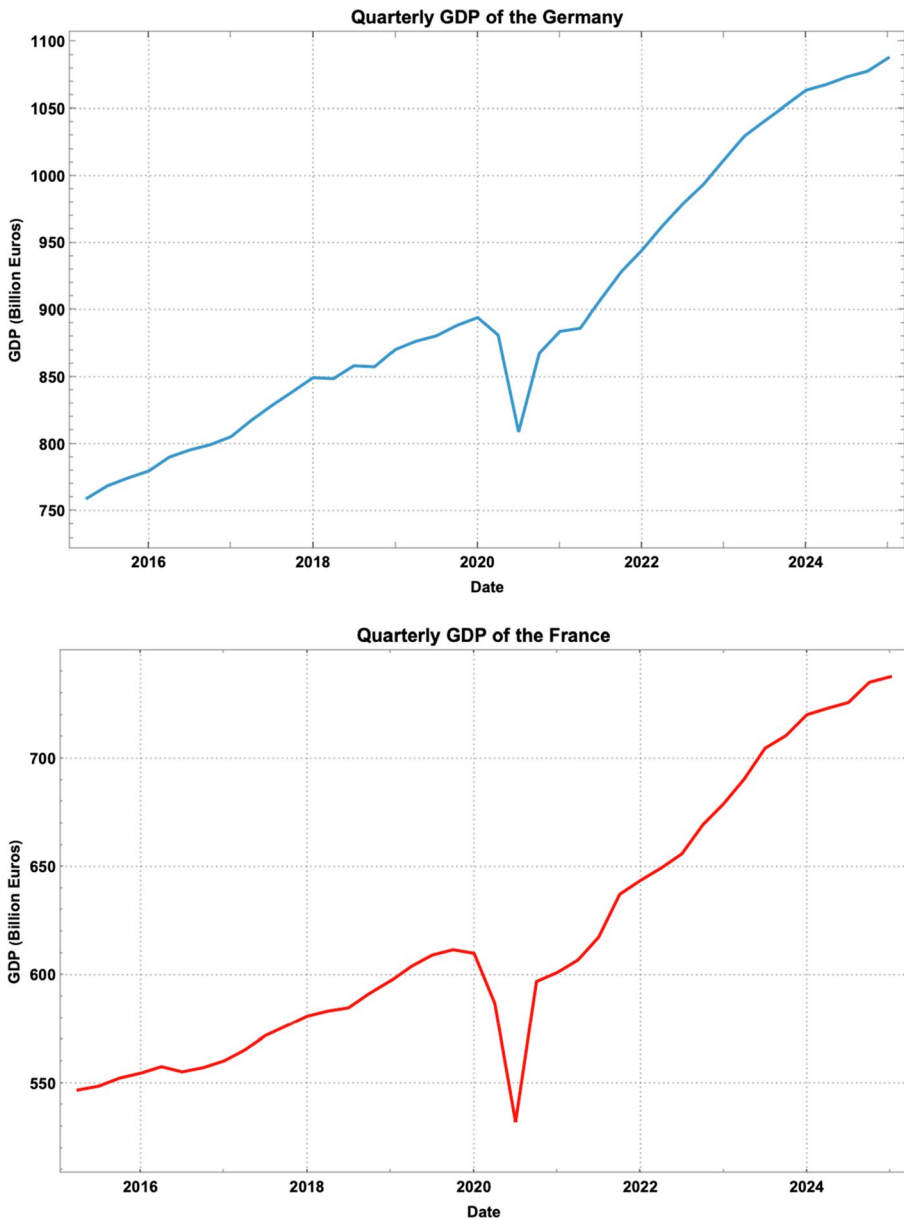


Fig. 4 Time series plot showing the long-run growth trajectory of GDP of the top 6 European economies. Note. The graph depicts a consistent upward trend in GDP, with a notable downturn reflecting the impact of a major economic disruption. The steady growth observed in later years indicates recovery and continued expansion

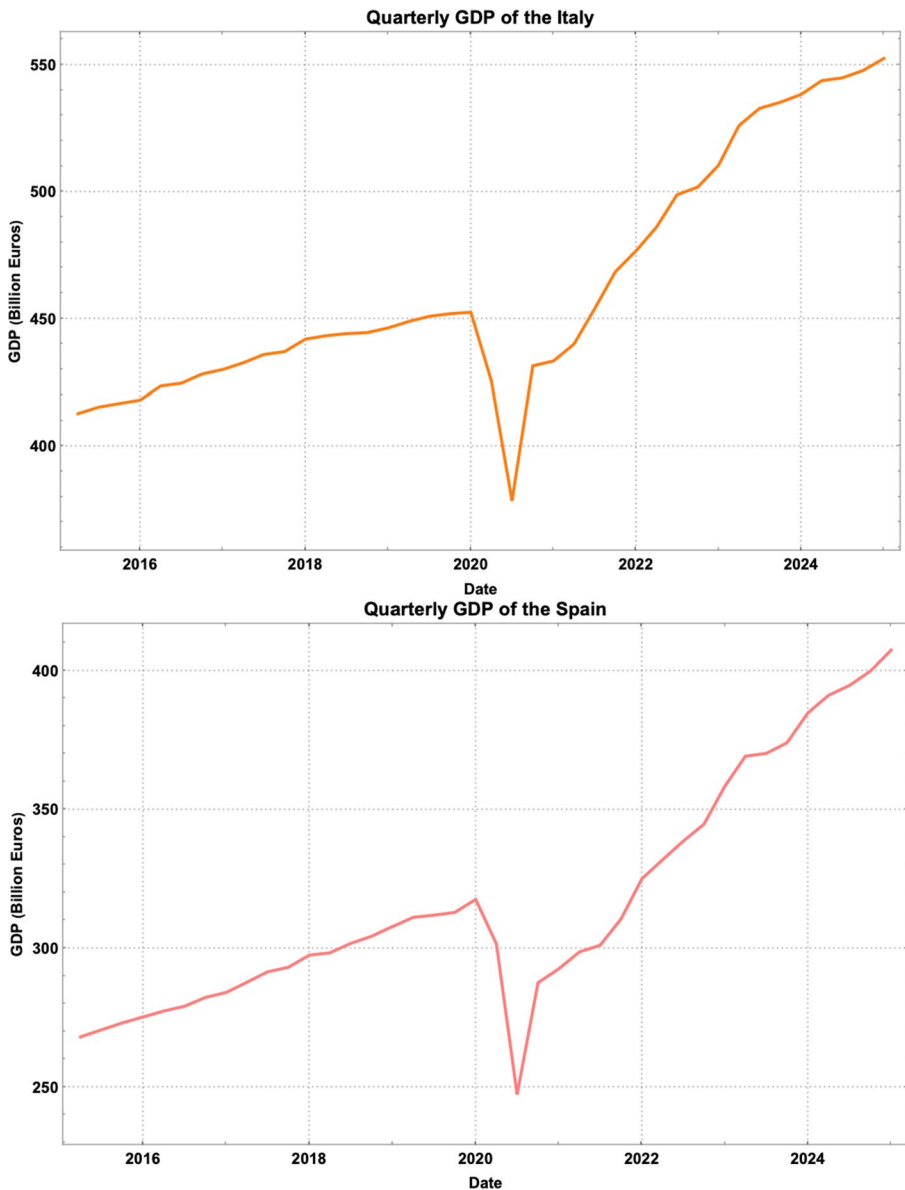


Fig. 4 (continued)

that variations in electricity use do not consistently lead to or forecast economic growth. Yet, GDP growth does seem to Granger-cause electricity use in Germany and Italy—namely, at higher lags—implying economic growth could lead to higher energy consumption in these economies. There is weak evidence in France of this unidirectional causality. Spain, the Netherlands, and Switzerland, by contrast, demonstrate no substantial causality in either direction. These results confirm economic activity can

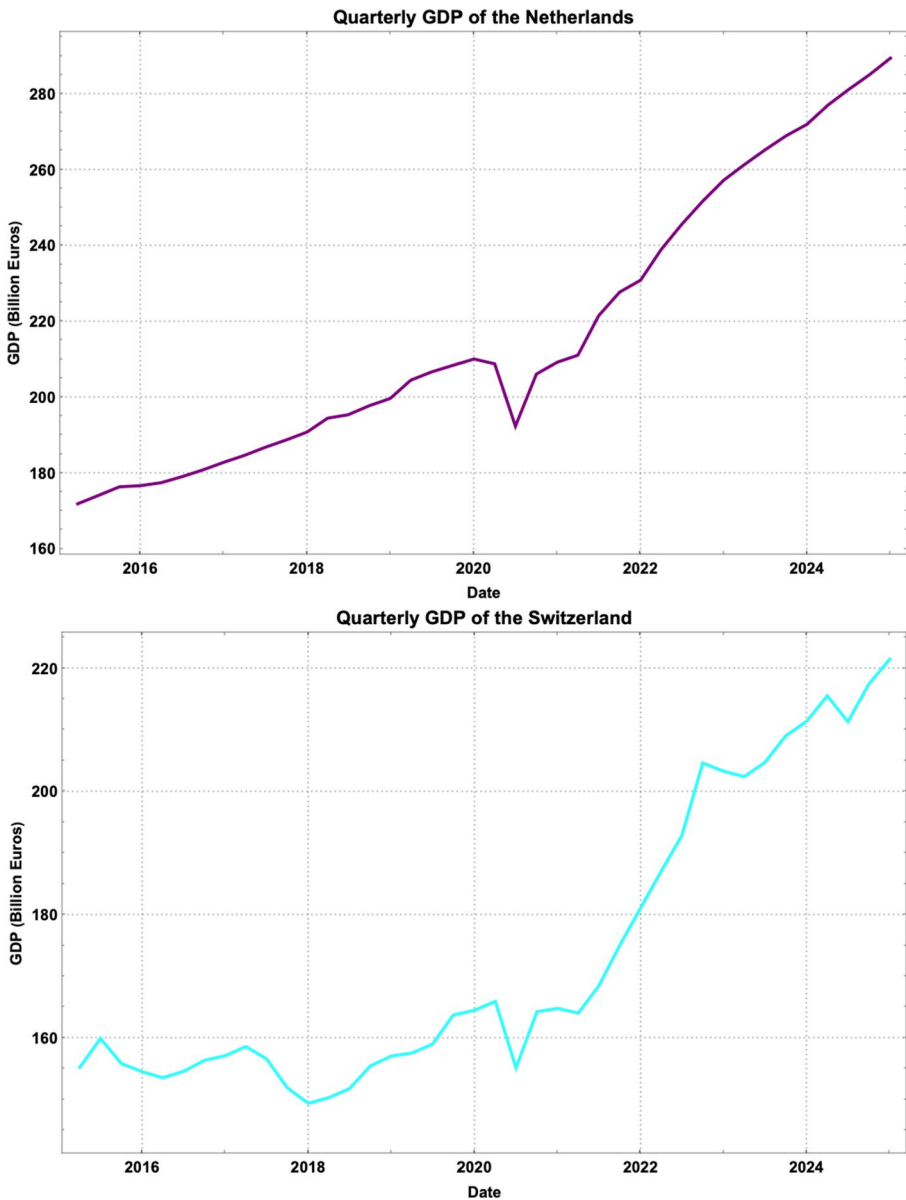


Fig. 4 (continued)

be a more potent source of energy demand, not the other way around, particularly in more efficient or advanced economies (Table 2).

The VAR evidence reveals heterogeneous dynamic links between electricity consumption and GDP growth across the six European countries of focus. For Germany, Italy, and France, evidence indicates GDP growth has a strong impact on electricity consumption, while Germany and Italy show unidirectional causality from GDP to

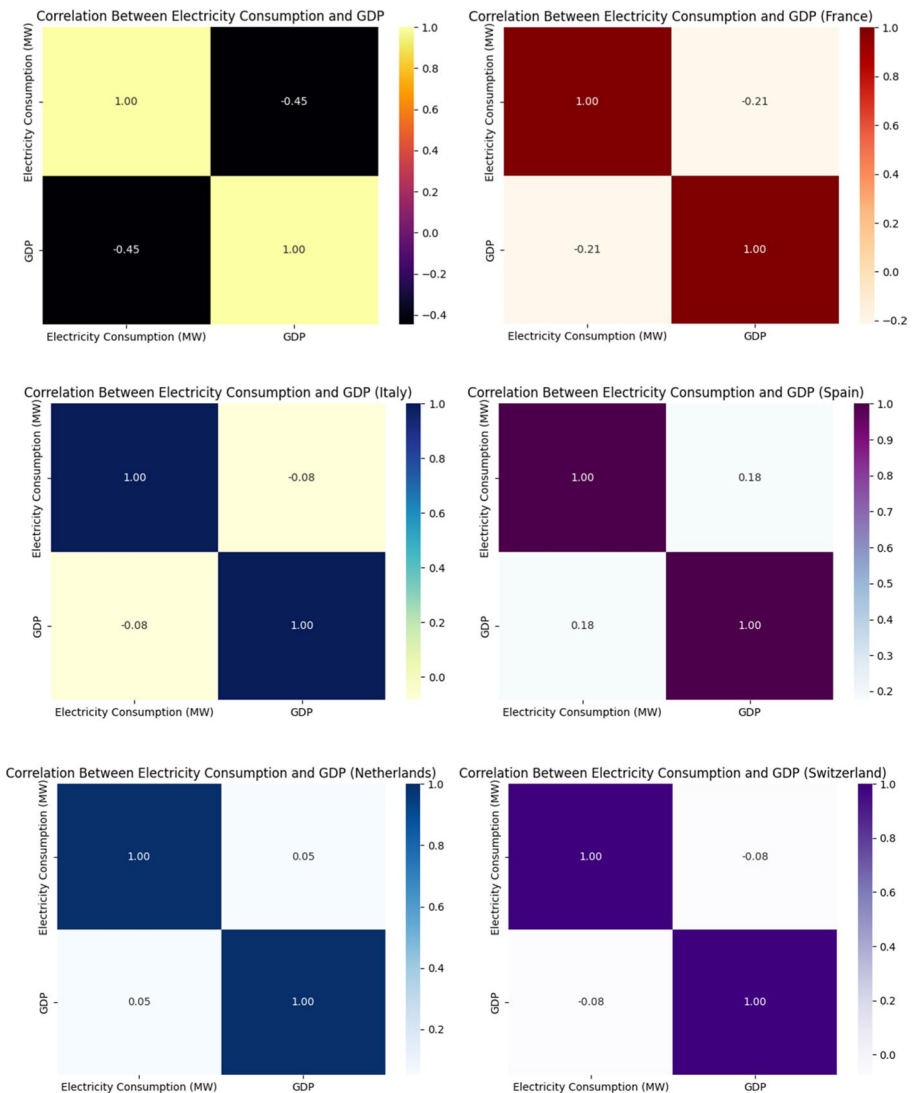


Fig. 5 Correlation matrix between electricity consumption and GDP of top 6 European economies. Note: This correlation matrix shows a very weak positive correlation (0.05) between electricity consumption and GDP in the Netherlands, indicating limited linear association during the observed period

electricity, but bidirectional causality in the case of France. For Spain, the Netherlands, and Switzerland, no important causal links are revealed, meaning economic growth and electricity use in these nations follow mostly independently. The evidence is further supported by IRFs and FEVD, showing larger interdependencies in economically industry-driven economies. The results overall illustrate that the electricity-GDP nexus is non-uniform across Europe, highlighting the necessity of country-specific economic and energy policies (Table 3).

Table 1 Stationarity test results for GDP and electricity consumption across selected European economies

Country	GDP (Level)	Electricity (Level)	log(GDP) Diff.	log(Electricity) Diff.	log(Electricity) 2nd Diff.
Germany	Non-stationary	Non-stationary	Stationary	Stationary	-
France	Non-stationary	Non-stationary	Stationary	Stationary	-
Italy	Non-stationary	Stationary	Stationary	Stationary	-
Spain	Non-stationary	Stationary	Stationary	Non-stationary	Stationary
Netherlands	Non-stationary	Non-stationary	Stationary	Non-stationary	Stationary
Switzerland	Non-stationary	Non-stationary	Stationary	Non-stationary	Stationary

The table summarizes the stationarity properties of the GDP and electricity consumption series at different transformations. Tests were performed using the ADF method. Log-differencing was applied to achieve stationarity where necessary, with second differencing used in cases of persistent non-stationarity

Table 2 Granger causality test results between electricity consumption and GDP in selected European countries

Country	Electricity → GDP	GDP → Electricity
Germany	No	Yes (lag 4)
France	No	Weak (lag 4)
Italy	No	Yes (lag 2,4)
Spain	No	No
Netherlands	No	No
Switzerland	No	No

The table presents the direction and significance of Granger causality between electricity consumption and GDP. Causality is tested using optimal lag lengths selected by AIC. Results indicate unidirectional causality from GDP to electricity in Germany, France, and Italy, with no significant causality detected in the remaining countries

Table 3 VAR model Estimation results for electricity consumption and GDP across selected European economies

Country	Se-lected Lag (AIC)	Significant Effects ($p < 0.05$)	Residual Correlation	AIC	FPE
Germany	4	Electricity_Growth→Electricity_Growth (L2, L4), GDP_Growth→Electricity_Growth (L4)	0.4045	-14.71	4.17e-07
France	4	Electricity_Growth→Electricity_Growth (L1, L2, L3)	0.47	-13.08	2.14e-06
Italy	4	Electricity_Growth→Electricity_Growth (L4), GDP_Growth→Electricity_Growth (L4)	0.7145	-13.46	1.45e-06
Spain	0	None	0.0265	-8.12	0.000298
Netherlands	3	Electricity_Consumption→Electricity_Consumption (L1, L2, L3)	0.3579	-13.61	1.24e-06
Switzerland	4	Electricity_Consumption→Electricity_Consumption (L1, L2, L3, L4)	0.3096	-13.51	1.39e-06

The table summarizes key results from the VAR model, including optimal lag length (based on AIC), significant lagged effects ($p < 0.05$), residual correlation, Akaike Information Criterion (AIC), and Final Prediction Error (FPE). Results vary across countries, with significant autoregressive effects observed in electricity growth but limited cross-variable interactions

According to the IRFs and FEVD findings in Germany, France, Italy, Spain, the Netherlands, and Switzerland, the dynamic correspondence between electricity consumption and GDP growth reveals country-specific trends. For most countries, including Germany, France, and Italy, the IRFs show the effect of electricity consumption shocks to be relatively weak and statistically insignificant for GDP growth, implying a loose short-run causality. In contrast, GDP growth does exercise a more direct and sometimes significant influence on electricity consumption, especially in Italy and the Netherlands. FEVD results support this observation that the GDP growth is explained mainly by its innovations (over 95%), while a substantial share of the forecast error variance in electricity consumption (~15% up to more than 50%) arises from the shocks in GDP growth. The implications are that electricity consumption does not appear to be a robust driver of GDP in the short term, whereas fluctuations in GDP can drive electricity consumption, highlighting the role of economic activity in determining energy demand as opposed to vice versa in these advanced European countries (Figs. 6 and 7).

Initial empirical testing relying on conventional econometric methodologies-the ADF test, Granger causality test, VAR, IRFs, and FEVD-failed to produce strong evidence of a persistent or large relationship between electricity consumption and GDP growth in the target European countries. The ADF test confirmed stationarity upon differencing, showing the occurrence of non-stationarity in the original series. Furthermore, tests for Granger causality were unable to find a definite causality in most cases. The VAR models demonstrated weak significance of coefficients, while plots of IRFs revealed short or marginal responses. The FEVD also revealed that every variable was mostly impacted by its shock, with no substantial contributions from the rest. This implies that linear models might not completely represent the latent dynamics of the electricity-GDP relationship. Therefore, in consideration of possible nonlinear trends and intricate temporal dependencies, this research applies an LSTM neural network, a deep learning model, highly efficient in sequence prediction and time series modeling.

The training performance of LSTM neural networks for the prediction of GDP growth based on electricity consumption in six European countries is shown in Table 4. Each country was represented by five distinct neural network architectures (from 1NN to 5NN) with differing activation function configurations. On the whole, the models demonstrated highly effective predictive performance, a range of around 0.9647 to 0.9998, signifying superior nonlinear learning. From Germany, the best performance was extracted from the model of 4NN (Ramp, Sin, ArcTan) as 0.9990. Italy and Spain also painted a better-than-average picture with performances of 0.9998 and 0.9991, respectively, from their 4NN and 5NN models, indicating electricity consumption is a highly effective indicator of GDP in these two nations when non-linearly modeled. France and the Netherlands also performed well, though slightly lower, with the best model positions in the range from around 1NN-4NN. In contrast, though a step or two behind the rest, Switzerland's best model (5NN) was also highly accurate in its performance at 0.9988. The use of activation functions such as Ramp, Sin, and ArcTan, when used together, seems effective in capturing complex data, with

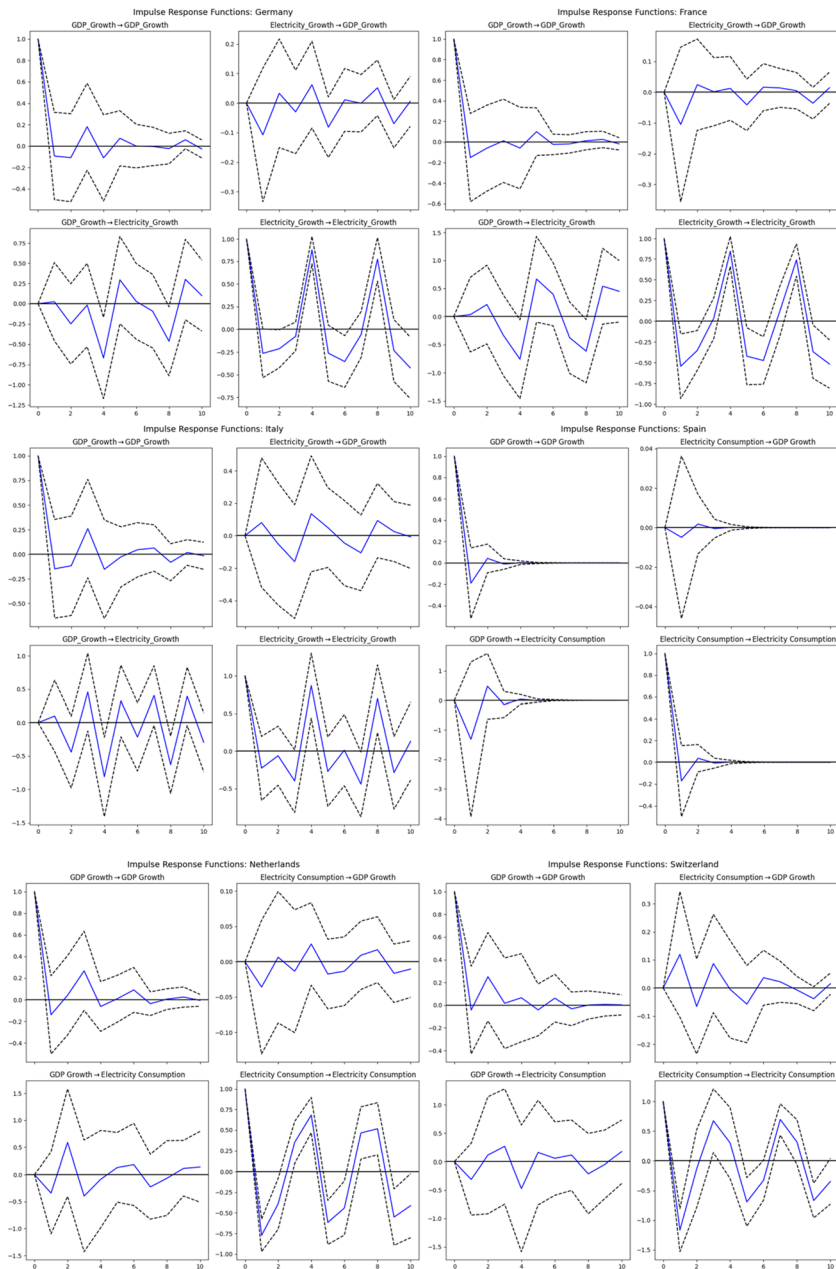
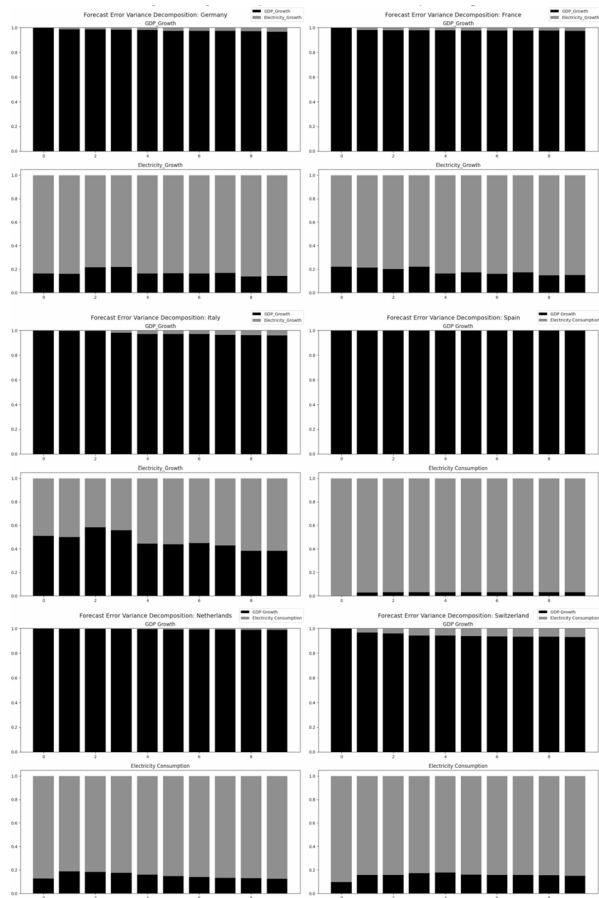


Fig. 6 IRFs of the top 6 European economies: GDP Growth and Electricity Consumption Dynamics. Note: This figure presents the IRFs from the VAR model for Switzerland, showing the response of GDP growth and electricity consumption to one-standard-deviation shocks in each variable. The solid blue lines represent the responses, while the dashed lines indicate 95% confidence intervals. The plots suggest weak and short-lived cross-variable effects, with stronger autoregressive dynamics observed in electricity consumption

Fig. 7 FEVD of GDP Growth and Electricity Consumption of the top 6 European economies. Note: The figure shows the FEVD results for Switzerland, illustrating the proportion of forecast error variance in GDP growth and electricity consumption attributed to shocks in each variable over a 10-period horizon. GDP forecast variance is primarily explained by its shocks, while electricity consumption variance shows a modest contribution from GDP shocks, indicating limited cross-variable predictive power



the use of Ramp commonly in the first layer across the top-performing models. The evidence identifies the superiority of the LSTM model in modeling the electricity-GDP relationship when compared to traditional methods in econometrics.

The graphs comparing actual vs. predicted GDP values for all six nations exhibit a remarkably close correspondence, showing that the LSTM neural network models captured the underlying patterns of the data. For all cases, the predicted GDP path traces the actual values closely, with tiny deviations, even during economic crises or fluctuations. This feature speaks of the robust forecasting power of the LSTM models, implying that electricity consumption is a sound input feature for GDP forecasting when predicted through deep learning methods. The results also reflect the strength of LSTM in modeling temporal dependency, which might be missed by traditional econometric models (Fig. 8).

The descriptive statistics reveal that the LSTM neural network model successfully approximates real GDP values for all six countries. For every country, the forecasted GDP closely reflects the real GDP in terms of mean, minimum, maximum, and

Table 4 Neural network performance and activation functions for GDP forecasting across six European economies

(a) Germany				
Network	Performance	Activation Function 1	Activation Function 2	Activation Function 3
1NN	0.998924	Ramp	Sin	Sin
2NN	0.995443	Ramp	Sin	Ramp
3NN	0.998179	Ramp	Sin	Sin
4NN	0.999013	Ramp	Sin	ArcTan
5NN	0.994086	Sin	Sin	Sin
(b) France				
Network	Performance	Activation Function 1	Activation Function 2	Activation Function 3
1NN	0.991971	Ramp	ArcTan	ArcTan
2NN	0.989299	Ramp	Ramp	ArcTan
3NN	0.964668	ArcTan	Ramp	ArcTan
4NN	0.968051	Sin	ArcTan	ArcTan
5NN	0.968026	ArcTan	ArcTan	ArcTan
(c) Italy				
Network	Performance	Activation Function 1	Activation Function 2	Activation Function 3
1NN	0.998285	Sin	ArcTan	ArcTan
2NN	0.999587	ArcTan	Ramp	Ramp
3NN	0.999464	Ramp	ArcTan	Ramp
4NN	0.99984	Ramp	Sin	Sin
5NN	0.997662	ArcTan	ArcTan	ArcTan
(d) Spain				
Network	Performance	Activation Function 1	Activation Function 2	Activation Function 3
1NN	0.992518	Sin	Ramp	Ramp
2NN	0.99653	Ramp	Ramp	Sin
3NN	0.992714	Ramp	Sin	Ramp
4NN	0.998457	Ramp	Ramp	Ramp
5NN	0.999133	Ramp	Ramp	Ramp
(e) Netherlands				
Network	Performance	Activation Function 1	Activation Function 2	Activation Function 3
1NN	0.995746	Ramp	Sin	Sin
2NN	0.986241	Sin	Sin	Ramp
3NN	0.993916	Ramp	Ramp	Sin
4NN	0.99748	Ramp	Ramp	Sin
5NN	0.986241	Sin	Sin	Ramp
(f) Switzerland				
Network	Performance	Activation Function 1	Activation Function 2	Activation Function 3
1NN	0.98678	Ramp	Ramp	Ramp
2NN	0.995805	Ramp	Ramp	Sin
3NN	0.998337	Sin	Ramp	Ramp
4NN	0.985867	Ramp	Sin	Sin
5NN	0.998824	Sin	Ramp	Ramp

This table presents the performance (R^2 values) of five neural network (NN) configurations tested for each country, along with the activation functions used in each of the three layers. The best-performing configurations—often combining Ramp, Sin, and ArcTan functions—demonstrate high predictive accuracy ($R^2 > 0.99$ in several cases), highlighting the effectiveness of neural networks in capturing complex nonlinear patterns between electricity consumption and GDP

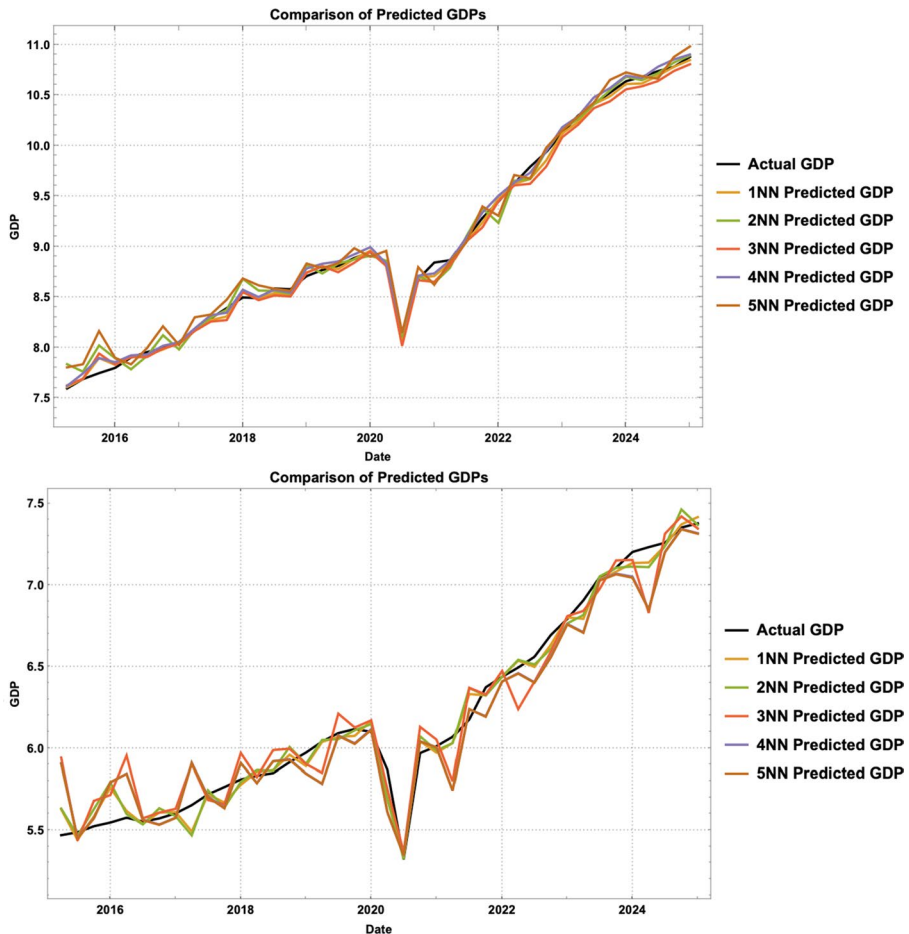


Fig. 8 Comparison of Neural Network Forecasts with Actual GDP Values of the top 6 European economies (Germany, France, Italy, Spain, the Netherlands, and Switzerland). Note: The graphs show a close match between actual and predicted GDP values, highlighting the strong predictive performance of the LSTM models. This supports the potential of electricity consumption data in forecasting economic growth

dispersion indicators (variance, standard deviation, and deviations). The slight variations in quartile, mean, and median deviations also show the model not only captures central tendencies but preserves the structure in the data distribution, thus validating the model's accuracy and stability in projecting GDP from electricity consumption trends (Table 5).

The comparison between actual and predicted GDP values demonstrates the high accuracy of the LSTM neural network models in forecasting economic growth using electricity consumption data. The predicted values closely follow the actual GDP trends across Germany, France, Italy, Spain, the Netherlands, and Switzerland, highlighting the model's robustness in capturing temporal patterns (Fig. 9).

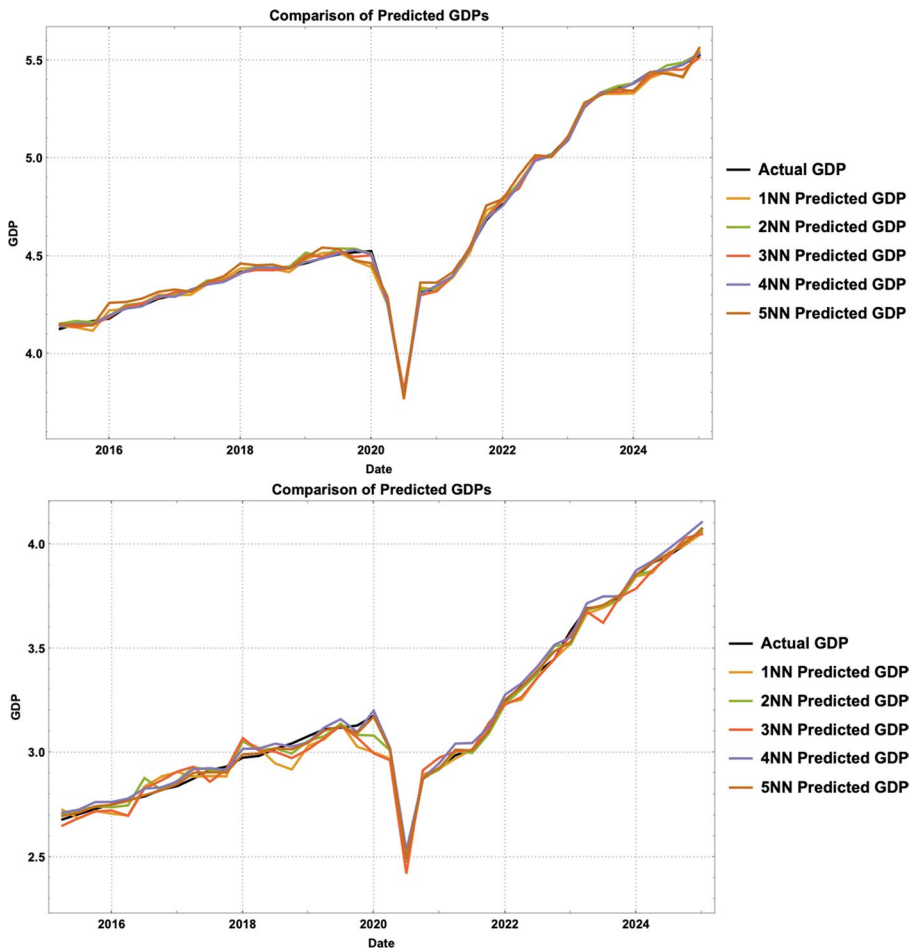


Fig. 8 (continued)

The error metrics RMSE, MAE, Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2) demonstrate the highly superior performance of the LSTM neural network model in forecasting GDP from electricity consumption for all six countries. For Germany, the low average prediction errors are indicated by an RMSE of 5.114 and an MAE of 4.149, along with a highly impressive R^2 of 0.997, suggesting that the model captures nearly all the changes in GDP. The same pattern applies to France, where performance remains strong but with slightly larger errors (RMSE = 7.708, MAE = 5.726), and a good R^2 of 0.984. Italy shows the best overall performance, with an RMSE of just 0.821 and an MAPE of only 0.147%, and a near-perfect R^2 of 0.999, indicating highly accurate predictions. Spain follows closely with an R^2 of 0.998 and low error values, supporting high reliability. The Netherlands also performs well, with an R^2 of 0.995, while Switzerland achieves an R^2 of 0.997,

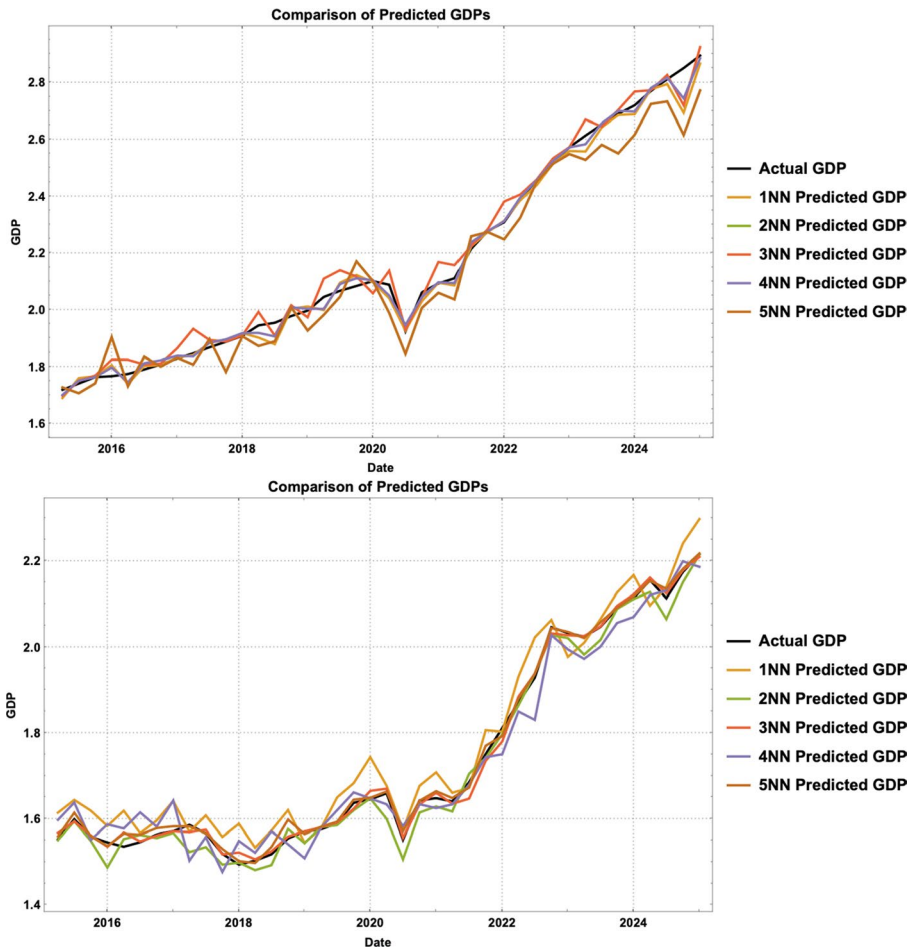


Fig. 8 (continued)

along with lower MAPE percentages of 0.849% and 0.593%, respectively, reflecting the success of the LSTM approach. Overall, these metrics showcase the model's robustness and reliability across countries with different levels of GDP, confirming its suitability for capturing complex, nonlinear relationships between electricity consumption and economic output (Table 6).

Walk-Forward Forecasting Results In the Table 7 shows the walk-forward forecasting performance by using the LSTM model across the top economic countries in Europe. As we can see that the difference between full-sample training and the walk-forward forecasting performance is evident by looking at the R^2 values. Spain (-3.676), Germany (-1.636), and Switzerland (-2.368) showed poor performance in the R^2 values,

Table 5 Descriptive statistics of actual and LSTM-Predicted GDP for selected European countries

Country		Germany		France	
Description		Actual GDP	GDP - LSTM Neural Network Prediction	Actual GDP	GDP - LSTM Neural Network Prediction
Minimum		759.537	762.032	532.04	535.041
Maximum		1087.97	1090.3	737.811	741.68
Mean		903.455	906.067	618.62	618.09
Variance		10036.3	10053.5	3751.74	3595.28
Standard Deviation		100.181	100.267	61.2514	59.9607
Quartile Deviation		81.5427	79.1389	47.05	45.2676
Median Deviation		73.877	72.8901	43.6446	41.2236
Mean Deviation		83.5621	83.7518	51.1496	50.3285
Country		Italy		Spain	
Description		Actual GDP	GDP - LSTM Neural Network Prediction	Actual GDP	GDP - LSTM Neural Network Prediction
Minimum		378.51	378.239	247.347	248.899
Maximum		552.552	553.482	407.457	407.325
Mean		463.182	463.174	316.897	316.964
Variance		2154.54	2150.43	1724.39	1720.45
Standard Deviation		46.417	46.3727	41.5258	41.4783
Quartile Deviation		34.8057	34.8986	26.982	27.5037
Median Deviation		21.3471	21.8955	21.3895	20.9792
Mean Deviation		38.9937	38.9251	33.5622	33.6597
Country		Netherlands		Switzerland	
Description		Actual GDP	GDP - LSTM Neural Network Prediction	Actual GDP	GDP - LSTM Neural Network Prediction
Minimum		171.938	170.158	149.401	149.774
Maximum		289.478	288.535	221.507	221.776
Mean		217.251	216.984	173.592	174.269
Variance		1308.93	1267.59	548.751	540.54
Standard Deviation		36.1791	35.6033	23.4254	23.2495
Quartile Deviation		30.4648	30.0144	20.9978	20.6808
Median Deviation		24.0093	23.7952	9.3236	8.40701
Mean Deviation		30.7778	30.4747	20.3686	20.2047

The table compares actual GDP values with those predicted by the LSTM model, showing close alignment in central tendency and dispersion measures, indicating high model accuracy

while the Netherlands showed positive performance, but that's not enough and not even close to the LSTM prediction of the full dataset. These findings point out that electricity consumption alone does not robustly capture short-term GDP variation under strict validation settings. Overall, the results align with the notion of electricity-GDP decoupling in post-industrial economies and highlight the need for additional predictors or hybrid models for effective forecasting.

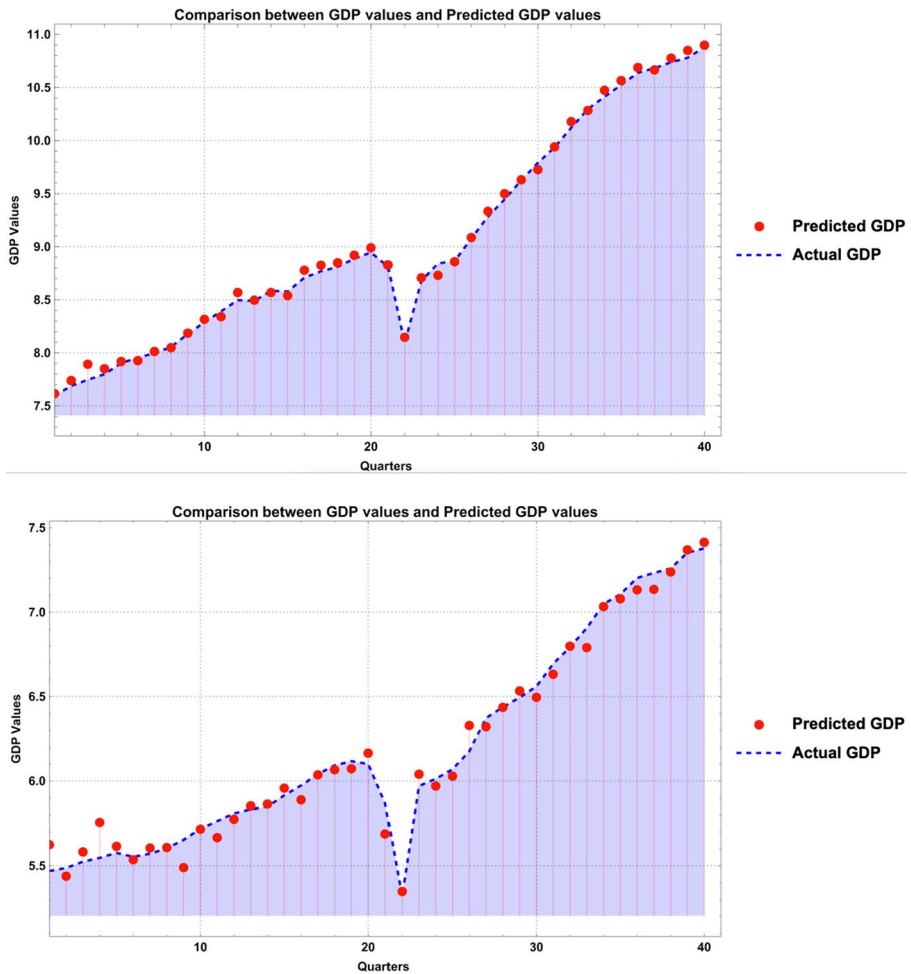


Fig. 9 Actual vs. Predicted GDP Values Using Neural Network Model in the top 6 European economies (Germany, France, Italy, Spain, the Netherlands, and Switzerland). Note: The graphs illustrate a close alignment between actual and predicted GDP values, indicating that the LSTM model successfully captured the underlying growth patterns across all six countries

4 Discussion of Results

RQ1: To what extent does electricity consumption predict GDP growth in major European economies, and how effectively can machine learning models capture this relationship compared to traditional econometric approaches?

The empirical exploration of the nexus between electricity usage and GDP in key European economies reveals divergent findings from traditional econometric methods and neural network-driven ones. First, the study familiarized conventional time series econometric methods like the ADF test, the Granger causality, and VAR models

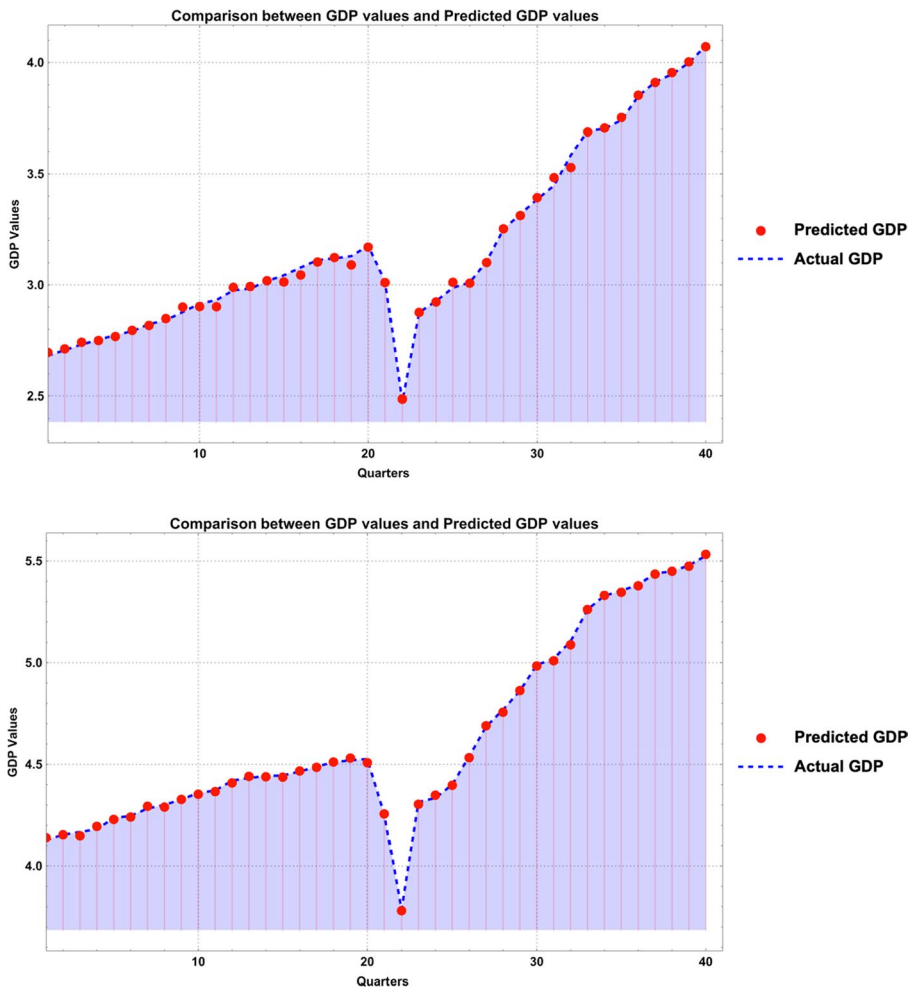


Fig. 9 (continued)

in analyzing causality and co-movement between economic activity and electricity usage. The ADF test revealed that the original data for both economic activity (GDP) and electricity usage were not stationary, but the same was realized after differencing. The findings of the Granger causality, however, were generally indeterminate across the countries, exhibiting weak or no directional causality. The same was also the situation with VARs as well as the later IRFs and FEVD, with these yielding country-specific but minimal results in many cases, frequently unable even to prove a statistically significant or stable causal effect of electricity usage on economic activity.

Due to the low explanatory content of traditional models, the study changed the focus towards machine learning approaches with LSTM neural networks. Compared to lag-based linear-assumption econometric methods, LSTM networks are capable of modeling complex, nonlinear temporal dependencies in high-frequency data. The performance metrics of the LSTM models, including high R^2 values (often well

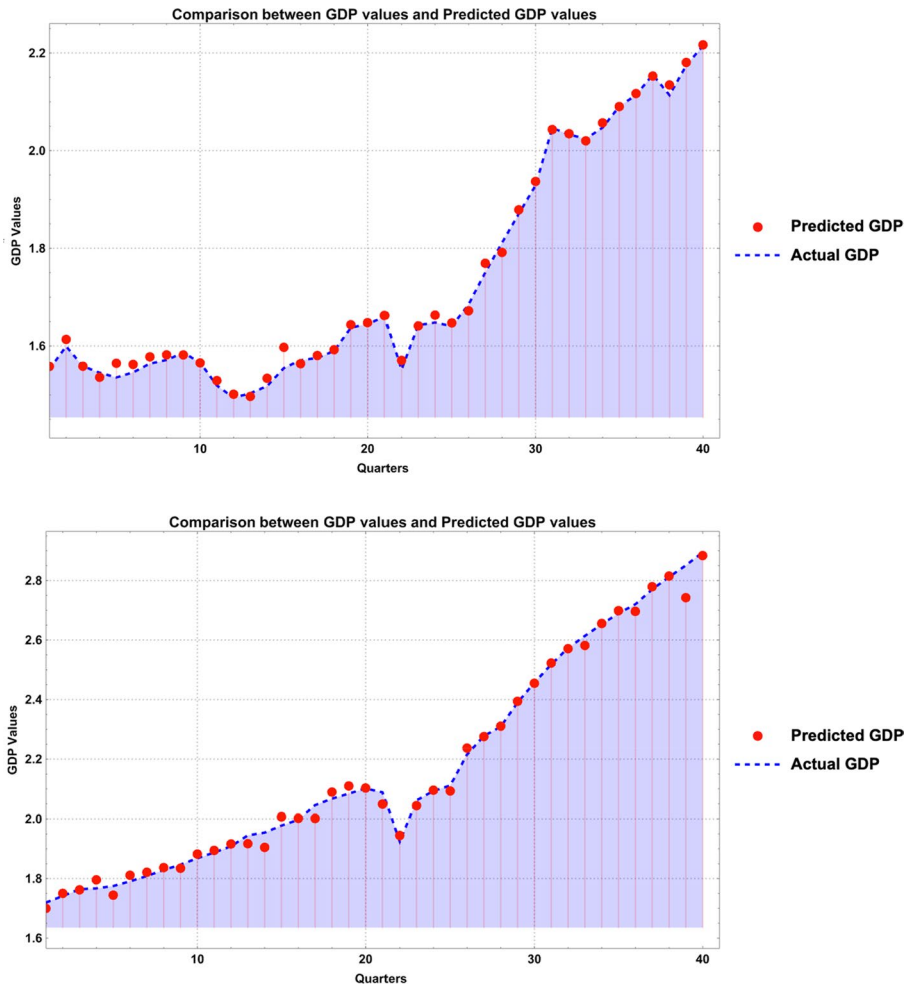


Fig. 9 (continued)

Table 6 Error metrics for LSTM-Based GDP prediction across six European countries

	RMSE	MAE	MAPE(%)	R^2
Germany	5.114	4.149	0.466	0.997
France	7.708	5.726	0.949	0.984
Italy	0.821	0.663	0.147	0.999
Spain	1.708	1.193	0.383	0.998
Netherlands	2.593	1.808	0.849	0.995
Switzerland	1.318	0.985	0.593	0.997

RMSE, MAE, MAPE, and R^2 values indicate high prediction accuracy of the LSTM model, with all countries showing R^2 values above 0.98, confirming strong model performance in estimating GDP from electricity consumption

Table 7 Walk-Forward forecasting performance of LSTM models for GDP prediction using electricity consumption

	RMSE	MAE	MAPE(%)	
Germany	1.340	1.184	12.18	-1.636
France	0.911	0.684	10.47	-1.579
Italy	0.603	0.497	9.96	-0.483
Spain	0.948	0.611	19.71	-3.676
Netherlands	0.272	0.213	8.84	0.140
Switzerland	0.388	0.343	17.97	-2.368

This table reports the RMSE, MAE, MAPE, and R^2 for each country under a walk-forward validation strategy. Negative R^2 values indicate that the model performed worse than a naive mean-based predictor, reflecting limited generalizability of electricity consumption for short-term GDP forecasting in most cases

above 0.99), very low RMSE, MAE, and MAPE percentages, showed their higher ability to predict GDP from electricity consumption for all six countries. Besides, graphical representations of predicted vs. actual GDP values showed close agreement. While previous models failed to uncover substantial structural relationships, the persistent accuracy of LSTM projections implies that latent economic patterns exist in electricity consumption. These include seasonal industrial swings, cyclical demand shifts, and sector-specific consumption patterns that reflect short-term economic activity. LSTM networks are designed to learn such nonlinear and temporal correlations without explicit feature engineering. This allows them to identify tiny changes in power demand that correspond to post-pandemic recoveries, supply chain rebounds, and sectoral rebalancing, even in decoupled economies. This is indicative that the consumption of electricity does contain significant predictive information for GDP, yet it requires advanced, non-linear modeling techniques to discern its impact. Laborda et al. (2023) adopted Temporal Fusion Transformers (TFTs) to forecast GDP for 25 OECD countries, achieving large improvements in RMSE and MAE compared with traditional ARIMA models, particularly in times of low economic volatility. While our LSTM-based model, as a basic, provided incredibly high accuracy when applied to most countries, with a comparable RMSE of 0.82 (Italy) and $R^2 > 0.98$ for the majority. This comparison emphasized that LSTM, without a more sophisticated attention mechanism, can still be used as an effective and scalable forecasting tool, especially for decoupled high-income economies.

Electricity consumption reflects real-time industrial, commercial, and household activity, which often follows seasonal and cyclical patterns. While no longer causally linked to GDP in high-income countries, these patterns embed signals of short-term economic fluctuations. LSTM models effectively capture such nonlinear dependencies, especially amid structural shifts like post-COVID recovery or improved energy efficiency. Our results reinforce existing literature on electricity-GDP decoupling in high-income economies, as evident in Liddle et al. (2023), who report declining elasticity between electricity demand and GDP within the OECD. The lack of Granger causality and low correlation in countries such as Germany and Switzerland aligns with these findings. This study contributes by showing that, although structural causality is absent, electricity data remains a useful proxy for near-term GDP forecasting when interpreted through nonlinear models like LSTM.

RQ2: How does the predictive performance of machine learning models, specifically LSTM neural networks, differ from traditional econometric models in capturing the dynamics between electricity consumption and GDP in top European economies?

The cross-sectional comparison between traditional econometric modeled outcomes and machine learning neural network modeled outcomes, namely, LSTM modeled results, reveals a wide difference in predictive strength and interpretive value in analyzing the electricity-GDP relationship. The traditional VAR-modeled results, along with the traditional Granger causality, were originally employed in determining if any dynamic dependence or linear causality existed. Results for Germany, however, were mostly statistically insignificant. The traditional Granger causality was often unable to specify directionality, while the IRF yielded small or transitory impacts of shocks in electricity on GDP. These results highlight a basic drawback of economic modeling: their linearity assumption, stationarity assumption, along preselected lag structures, not being capable of capturing complex, nonlinear real-world data patterns.

In contrast, LSTM neural networks, proposed specifically for learning long-term dependence structures and nonlinear associations, were highly effective. Performance metrics, such as R^2 values, were always above 0.98, while error metrics, such as RMSE, MAE, and MAPE, were low in every country. Also, correlation analysis as well as prediction-actual comparison plots showed near-perfect GDP trends, highlighting the power of LSTM models in forecasting macroeconomic variables from energy data. However, to test the true predictive capability of the LSTM models, we implemented a walk-forward validation strategy using a 20-quarter rolling training window. While in-sample metrics showed high accuracy, the out-of-sample results were mixed, with R^2 values negative in most countries (e.g., Germany: -1.64 , Spain: -3.68), suggesting poor generalizability. Only the Netherlands showed marginally positive R^2 . These findings indicate that while electricity consumption contains signals useful for short-term forecasting, relying on it exclusively leads to overfitting and weak real-world performance when not carefully validated. The advantage of these models is automatic learning of optimal lag structures as well as nonlinear interactions, obviating the necessity for hand-tuning, while presenting a data-driven solution, capable of fine-tuning across changing economic contexts.

Ultimately, the machine learning models did not just outperform traditional methods in prediction but also captured latent patterns that conventional econometric frameworks missed. This reinforces the notion that integrating advanced ML approaches can significantly enhance macroeconomic forecasting, especially in energy-influenced economies. The practical implications of this research are multifaceted and include a variety of industry- as well as policy-related fields. Electricity consumption data in real time, combined with nonlinear forecasting by LSTM, can provide several national statistical offices, energy regulators, and industrial planners with more timely indicators for short-term economic activity beyond quarterly official releases such as GDP. This would be especially helpful in energy-consuming industries, like manufacturing, logistics, and utilities, where demand forecasting drives production schedules, capacity

planning, and supply chain optimization. LSTM models are scalable and can be retrained or fine-tuned for different geographies, industries, or on a firm-level basis if there is sufficient high-frequency data available. The LSTM model's accuracy arises from its ability to exploit temporal signals embedded in electricity consumption patterns, which correlate with short-term economic fluctuations but do not imply underlying causality.

To enhance interpretability, we examine what the LSTM models might be capturing. Given their structure, these models are likely to identify recurring nonlinear patterns in electricity demand that relate to economic seasonality (e.g., winter heating surges, summer industrial slowdowns), post-pandemic recovery phases, or production cycle restarts after holidays or disruptions. Unlike static econometric models, LSTMs can adapt to such variations in real time, explaining their superior ability to project GDP trends that linear models may overlook. This interpretive layer aligns with previous research using explainable AI in macroeconomic forecasting, which shows that deep learning models not only predict effectively but also reveal economically significant relationships.

5 Conclusion

This study investigated the relationship between electricity consumption and GDP in six advanced European economies: Germany, France, Italy, Spain, the Netherlands, and Switzerland, using a dual-methodological lens that combined traditional econometric diagnostics with deep learning architectures. Preliminary findings from unit root testing, Granger causality, VAR modeling, impulse response functions, and forecast error variance decomposition revealed primarily weak or statistically insignificant causal links, reflecting the structural decoupling common in post-industrial, energy-efficient economies. These findings highlight linear models' declining explanatory effectiveness in capturing the essentially nonlinear, dynamic interaction between energy usage and macroeconomic performance. We used Long Short-Term Memory neural networks to overcome restrictions and obtain remarkable short-term predicting performance across all nations. RMSE values were as low as 0.82, and R^2 regularly exceeded 0.98. The greater alignment of anticipated and observed GDP demonstrates LSTM's ability to discover latent temporal correlations and nonlinear regularities that standard models miss. However, to test the generalizability of these models, we used a walk-forward validation with a rolling training window. The findings revealed that, despite high in-sample accuracy, out-of-sample predictive potential was restricted. Most nations' R^2 values were negative, such as -1.64 for Germany and -3.68 for Spain, showing poor performance in real-world conditions. Only the Netherlands had a somewhat favorable R^2 . This emphasizes the difficulty of overfitting and the significance of strong validation frameworks in data-driven forecasting. Finally, this study views electricity consumption not as a structural cause but as a beneficial high-frequency indicator for forecasting economic activity. The findings support the incorporation of data-driven, adaptive modeling tools, such as neural networks, into the empirical macroeconomic toolkit, particularly for forecasting in complex, decoupled economic contexts.

Appendix

Table 8 ADF unit root test results for GDP and electricity consumption of the top 6 European economies

ADF test results: Germany			
Variable	ADF Statistic	p-value	Stationary
GDP	0.1428	0.9689	No
Electricity Consumption	1.5865	0.9978	No
Differenced log(GDP)	-7.1052	4.07e-10	Yes
Differenced log(Electricity)	-3.0022	0.0347	Yes
ADF test results: France			
Variable	ADF Statistic	p-value	Stationary
GDP	0.0566	0.963	No
Electricity Consumption	-0.5797	0.8755	No
Differenced log(GDP)	-7.3545	9.86e-11	Yes
Differenced log(Electricity)	-3.4422	0.0096	Yes
ADF test results: Italy			
Variable	ADF Statistic	p-value	Stationary
GDP	-0.1361	0.9457	No
Electricity Consumption	-2.8922	0.0463	Yes
Differenced log(GDP)	-6.8467	1.7353e-09	Yes
Differenced log(Electricity)	-2.9759	0.0372	Yes
ADF test results: Spain			
Variable	ADF Statistic	p-value	Stationary
GDP	0.0712	0.964	No
Electricity Consumption	-4.3826	0.0003	Yes
Differenced log(GDP)	-7.2853	1.464e-10	Yes
Differenced log(Electricity)	-0.4025	0.9097	No
2nd Difference log(Electricity)	-7.6953	1.385e-11	Yes
ADF test results: Netherlands			
Variable	ADF Statistic	p-value	Stationary
GDP	1.4382	0.9973	No
Electricity Consumption	-2.5297	0.1084	No
Differenced log(GDP)	-6.9272	1.11e-09	Yes
Differenced log(Electricity)	-2.4851	0.1191	No
2nd Difference log(Electricity)	-4.5056	0.0002	Yes
ADF test results: Switzerland			
Variable	ADF Statistic	p-value	Stationary
GDP	1.2136	0.9961	No
Electricity Consumption	-1.5788	0.4942	No
Differenced log(GDP)	-5.9885	1.77e-07	Yes
Differenced log(Electricity)	-2.5629	0.1009	No
2nd Differenced log(Electricity)	-2.959	0.0389	Yes

The table reports the results of the ADF unit root test applied to GDP and electricity consumption data in Germany. Both variables are non-stationary at level but become stationary after first differencing in logarithmic form, supporting the use of VAR modeling on differenced series

Table 9 Granger causality test results between electricity consumption and GDP in Germany at various lag lengths

Granger Causality Germany			
Lag	Electricity → GDP (p-value)	GDP → Electricity (p-value)	Causality Direction (if any)
1	0.173	0.3671	None
2	0.3475	0.5082	None
3	0.3783	0.9637	None
4	0.5354	0.0497	GDP → Electricity
Granger Causality France			
Lag	Electricity → GDP (p-value)	GDP → Electricity (p-value)	Causality Direction (if any)
1	0.0785	0.1513	None
2	0.2426	0.1096	None
3	0.4112	0.5073	None
4	0.569	0.1345	None
Granger Causality Italy			
Lag	Electricity → GDP (p-value)	GDP → Electricity (p-value)	Causality Direction (if any)
1	0.4954	0.0875	None
2	0.6005	0.0216	GDP → Electricity
3	0.5	0.1266	None
4	0.6679	0.0251	GDP → Electricity
Granger Causality Spain			
Lag	Electricity → GDP (p-value)	GDP → Electricity (p-value)	Causality Direction (if any)
1	0.816	0.3324	None
2	0.7859	0.496	None
3	0.919	0.8543	None
4	0.9812	0.94	None
Granger Causality Netherlands			
Lag	Electricity → GDP (p-value)	GDP → Electricity (p-value)	Causality Direction (if any)
1	0.8622	0.5649	None
2	0.673	0.3045	None
3	0.7223	0.6141	None
4	0.7718	0.65	None
Granger Causality Switzerland			
Lag	Electricity → GDP (p-value)	GDP → Electricity (p-value)	Causality Direction (if any)
1	0.1453	0.4252	None
2	0.0925	0.8222	None
3	0.1887	0.5602	None
4	0.2429	0.5687	None

The table reports Granger causality test results between electricity consumption and GDP in Germany across lags 1 to 4. While no causality is detected at lower lags, a statistically significant causal relationship from GDP to electricity consumption is identified at lag 4 ($p = 0.0497$), suggesting unidirectional influence in the short run

Table 10 AR model Estimation summary and regression output for top 6 European countries

Germany				
Lag	AIC	BIC	FPE	HQIC
0	-12.80	-12.71	2.770e-06	-12.77
1	-12.70	-12.43	3.058e-06	-12.61
2	-13.76	-13.32	1.058e-06	-13.61
3	-14.04	-13.41	8.106e-07	-13.82
4	-14.71*	-13.91*	4.173e-07*	-14.44*
VAR Model Summary		Summary of Regression Results		
Model	VAR	Method	OLS	
No. of Equations	2.00000	BIC	-13.9132	
Nobs	35.0000	HQIC	-14.4370	
Log likelihood	176.154	FPE	4.17283e-07	
AIC	-14.7131	Det(Omega mle)	2.64035e-07	
Results for equation GDP Growth				
	coefficient	std. error	t-stat	prob
const	0.010466	0.005800	1.804	0.071
L1.GDP Growth	-0.092473	0.207829	-0.445	0.656
L1.Electricity Growth	-0.107555	0.115433	-0.932	0.351
L2.GDP Growth	-0.113728	0.207700	-0.548	0.584
L2.Electricity Growth	-0.005154	0.112582	-0.046	0.963
L3.GDP Growth	0.133724	0.203066	0.659	0.510
L3.Electricity Growth	-0.063165	0.114340	-0.552	0.581
L4.GDP Growth	-0.094110	0.201910	-0.466	0.641
L4.Electricity Growth	0.051619	0.113819	0.454	0.650
Results for equation Electricity Growth				
	coefficient	std. error	t-stat	prob
const	0.008427	0.006885	1.224	0.221
L1.GDP Growth	0.024329	0.246677	0.099	0.921
L1.Electricity Growth	-0.264775	0.137010	-1.933	0.053
L2.GDP Growth	-0.239724	0.246523	-0.972	0.331
L2.Electricity Growth	-0.282013	0.133626	-2.110	0.035
L3.GDP Growth	-0.095141	0.241023	-0.395	0.693
L3.Electricity Growth	-0.236109	0.135712	-1.740	0.082
L4.GDP Growth	-0.777051	0.239652	-3.242	0.001
L4.Electricity Growth	0.730020	0.135094	5.404	0.000
Correlation matrix of residuals				
	GDP Growth		Electricity Growth	
GDP Growth	1.000000		0.404513	
Electricity Growth	0.404513		1.000000	
France				
Lag	AIC	BIC	FPE	HQIC
0	-10.17	-10.08	3.839e-05	-10.14
1	-10.17	-9.900	3.845e-05	-10.07
2	-12.29	-11.84	4.626e-06	-12.13
3	-12.98	-12.36*	2.328e-06	-12.77
4	-13.08*	-12.28	2.143e-06*	-12.80*
VAR Model Summary		Summary of Regression Results		
Model	VAR	Method	OLS	

Table 10 (continued)

No. of Equations	2.00000	BIC	-12.2772	
Nobs	35.0000	HQIC	-12.8009	
Log likelihood	147.523	FPE	2.14270e-06	
AIC	-13.0771	Det(Omega mle)	1.35579e-06	
Results for equation GDP Growth				
	coefficient	std. error	t-stat	prob
const	0.009584	0.006991	1.371	0.170
L1.GDP Growth	-0.151397	0.218599	-0.693	0.489
L1.Electricity Growth	-0.104852	0.128049	-0.819	0.413
L2.GDP Growth	-0.079369	0.215403	-0.368	0.713
L2.Electricity Growth	-0.048497	0.122137	-0.397	0.691
L3.GDP Growth	0.014318	0.213103	0.067	0.946
L3.Electricity Growth	-0.066590	0.126706	-0.526	0.599
L4.GDP Growth	-0.082215	0.206491	-0.398	0.691
L4.Electricity Growth	-0.033662	0.120654	-0.279	0.780
Results for equation Electricity Growth				
	coefficient	std. error	t-stat	prob
const	0.001108	0.010835	0.102	0.919
L1.GDP Growth	0.034423	0.338814	0.102	0.919
L1.Electricity Growth	-0.542799	0.198467	-2.735	0.006
L2.GDP Growth	0.237585	0.333860	0.712	0.477
L2.Electricity Growth	-0.643671	0.189304	-3.400	0.001
L3.GDP Growth	-0.158384	0.330296	-0.480	0.632
L3.Electricity Growth	-0.480786	0.196385	-2.448	0.014
L4.GDP Growth	-0.796258	0.320047	-2.488	0.013
L4.Electricity Growth	0.351859	0.187006	1.882	0.060
Correlation matrix of residuals				
	GDP Growth		Electricity Growth	
GDP Growth	1.000000		0.470033	
Electricity Growth	0.470033		1.000000	
Italy				
Lag Order Selection				
Lag	AIC	BIC	FPE	HQIC
0	-12.05	-11.96	5.867e-06	-12.02
1	-13.00	-12.73*	2.273e-06	-12.90
2	-12.99	-12.55	2.292e-06	-12.84
3	-13.22	-12.60	1.836e-06	-13.00
4	-13.46*	-12.66	1.455e-06*	-13.19*
VAR Model Summary		Summary of Regression Results		
Model	VAR	Method	OLS	
No. of Equations	2.00000	BIC	-12.6643	
Nobs	35.0000	HQIC	-13.1881	
Log likelihood	154.298	FPE	1.45488e-06	
AIC	-13.4642	Det(Omega mle)	9.20570e-07	
Results for equation GDP Growth				
	coefficient	std. error	t-stat	prob
const	0.008826	0.007875	1.121	0.262
L1.GDP Growth	-0.147898	0.255974	-0.578	0.563

Table 10 (continued)

L1.Electricity Growth	0.080714	0.203543	0.397	0.692
L2.GDP Growth	-0.146965	0.269676	-0.545	0.586
L2.Electricity Growth	-0.020283	0.225647	-0.090	0.928
L3.GDP Growth	0.259666	0.266094	0.976	0.329
L3.Electricity Growth	-0.155048	0.205884	-0.753	0.451
L4.GDP Growth	-0.123422	0.275942	-0.447	0.655
L4.Electricity Growth	0.078775	0.206777	0.381	0.703
Results for equation Electricity Growth				
	coefficient	std. error	t-stat	prob
const	0.006322	0.008371	0.755	0.450
L1.GDP Growth	0.099975	0.272090	0.367	0.713
L1.Electricity Growth	-0.227511	0.216358	-1.052	0.293
L2.GDP Growth	-0.403737	0.286655	-1.408	0.159
L2.Electricity Growth	-0.121490	0.239854	-0.507	0.612
L3.GDP Growth	0.321283	0.282847	1.136	0.256
L3.Electricity Growth	-0.402321	0.218846	-1.838	0.066
L4.GDP Growth	-0.747610	0.293315	-2.549	0.011
L4.Electricity Growth	0.650353	0.219795	2.959	0.003
Correlation matrix of residuals				
	GDP Growth		Electricity Growth	
GDP Growth	1.000000		0.714521	
Electricity Growth	0.714521		1.000000	
Spain				
Lag Order Selection				
Lag	AIC	BIC	FPE	HQIC
0	-8.070*	-7.981*	0.0003126*	-8.040*
1	-7.931	-7.661	0.0003598	-7.839
2	-7.860	-7.411	0.0003876	-7.707
3	-7.865	-7.236	0.0003886	-7.651
4	-7.651	-6.843	0.0004879	-7.376
VAR Model Summary		Summary of Regression Results		
Model	VAR	Method	OLS	
No. of Equations	2.00000	BIC	-7.85947	
Nobs	37.0000	HQIC	-8.02860	
Log likelihood	51.2315	FPE	0.000297533	
AIC	-8.12070	Det(Omega mle)	0.000254577	
Results for equation GDP Growth				
	coefficient	std. error	t-stat	prob
const	0.012838	0.007587	1.692	0.091
L1.GDP Growth	-0.189529	0.168630	-1.124	0.261
L1.Electricity Consumption	-0.004924	0.020996	-0.234	0.815
Results for equation Electricity Consumption				
	coefficient	std. error	t-stat	prob
const	0.011705	0.060149	0.195	0.846
L1.GDP Growth	-1.314524	1.336886	-0.983	0.325
L1.Electricity Consumption	-0.173182	0.166458	-1.040	0.298
Correlation matrix of residuals				
	GDP Growth		Electricity Consumption	

Table 10 (continued)

GDP Growth	1.000000		0.026519	
Electricity Consumption	0.026519		1.000000	
Netherlands				
Lag Order Selection				
Lag	AIC	BIC	FPE	HQIC
0	-11.81	-11.72	7.444e-06	-11.78
1	-11.61	-11.34	9.048e-06	-11.52
2	-12.67	-12.22	3.159e-06	-12.52
3	-13.53*	-12.91*	1.341e-06*	-13.32*
4	-13.38	-12.57	1.587e-06	-13.10
VAR Model Summary		Summary of Regression Results		
Model	VAR	Method	OLS	
No. of Equations	2.00000	BIC	-12.9882	
Nobs	35.0000	HQIC	-13.3955	
Log likelihood	152.855	FPE	1.24128e-06	
AIC	-13.6103	Det(Omega mle)	8.61997e-07	
Results for equation GDP Growth				
	coefficient	std. error	t-stat	prob
const	0.011978	0.005908	2.027	0.043
L1.GDP Growth	-0.139605	0.184423	-0.757	0.449
L1.Electricity Consumption	-0.035792	0.048292	-0.741	0.459
L2.GDP Growth	0.009386	0.186986	0.050	0.960
L2.Electricity Consumption	-0.026238	0.030765	-0.853	0.394
L3.GDP Growth	0.286301	0.186917	1.532	0.126
L3.Electricity Consumption	-0.046384	0.049040	-0.946	0.344
Results for equation Electricity Consumption				
	coefficient	std. error	t-stat	prob
const	0.000437	0.012347	0.035	0.972
L1.GDP Growth	-0.341754	0.385438	-0.887	0.375
L1.Electricity Consumption	-0.773050	0.100929	-7.659	0.000
L2.GDP Growth	0.277274	0.390794	0.710	0.478
L2.Electricity Consumption	-1.000000	0.064298	-15.553	0.000
L3.GDP Growth	-0.231444	0.390650	-0.592	0.554
L3.Electricity Consumption	-0.710775	0.102491	-6.935	0.000
Correlation matrix of residuals				
	GDP Growth		Electricity Consumption	
GDP Growth	1.000000		0.357851	
Electricity Consumption	0.357851		1.000000	
Switzerland				
Lag Order Selection				
Lag	AIC	BIC	FPE	HQIC
0	-10.99	-10.90	1.684e-05	-10.96
1	-10.82	-10.56	1.992e-05	-10.73
2	-12.65	-12.20	3.237e-06	-12.49
3	-13.33	-12.70	1.652e-06	-13.11
4	-13.51*	-12.70*	1.394e-06*	-13.23*
VAR Model Summary		Summary of Regression Results		
Model	VAR			

Table 10 (continued)

Method	OLS			
No. of Equations	2.00000	BIC	-12.7007	
Nobs	34.0000	HQIC	-13.2332	
Log likelihood	151.162	FPE	1.39449e-06	
AIC	-13.5088	Det(Omega mle)	8.71838e-07	
Results for equation GDP Growth				
	coefficient	std. error	t-stat	prob
const	0.007597	0.005244	1.449	0.147
L1.GDP Growth	-0.043106	0.197071	-0.219	0.827
L1.Electricity Consumption	0.119543	0.114093	1.048	0.295
L2.GDP Growth	0.285469	0.202353	1.411	0.158
L2.Electricity Consumption	0.079098	0.106545	0.742	0.458
L3.GDP Growth	0.050403	0.197986	0.255	0.799
L3.Electricity Consumption	0.158497	0.116076	1.365	0.172
L4.GDP Growth	0.004519	0.184823	0.024	0.980
L4.Electricity Consumption	0.126614	0.109677	1.154	0.248
Results for equation Electricity Consumption				
	coefficient	std. error	t-stat	prob
const	0.007282	0.008558	0.851	0.395
L1.GDP Growth	-0.308934	0.321622	-0.961	0.337
L1.Electricity Consumption	-1.167529	0.186201	-6.270	0.000
L2.GDP Growth	-0.260190	0.330242	-0.788	0.431
L2.Electricity Consumption	-1.459933	0.173882	-8.396	0.000
L3.GDP Growth	0.015965	0.323115	0.049	0.961
L3.Electricity Consumption	-1.174062	0.189437	-6.198	0.000
L4.GDP Growth	-0.283109	0.301634	-0.939	0.348
L4.Electricity Consumption	-0.468059	0.178994	-2.615	0.009
Correlation matrix of residuals				
	GDP Growth		Electricity Consumption	
GDP Growth	1.000000		0.309559	
Electricity Consumption	0.309559		1.000000	

This table reports AR lag selection criteria and VAR model estimates for GDP and electricity consumption growth across six European countries. Asterisks (*) indicate optimal lag length. Coefficients, standard errors, t-stats, and p-values are shown. Residual correlations suggest a moderate link between the two variables

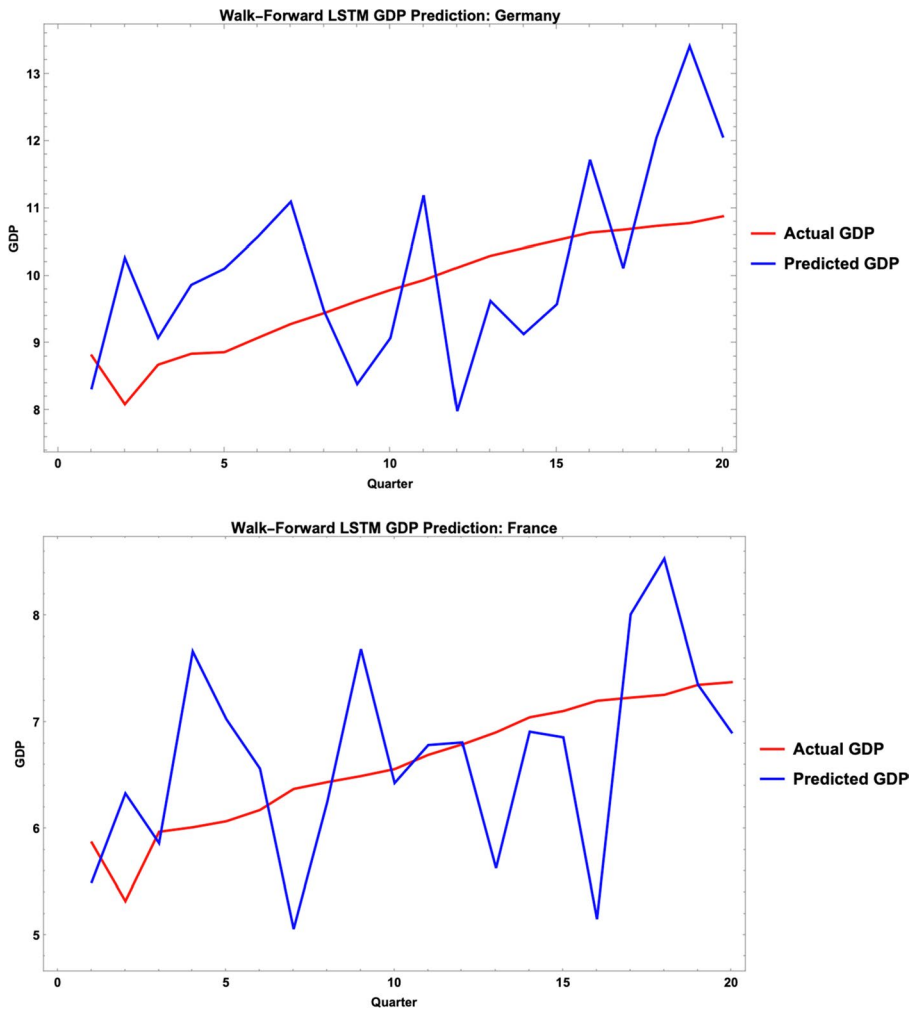


Fig. 10 Walk-Forward LSTM GDP Prediction for the top 6 European countries Based on Electricity Consumption. Note: The figure shows walk-forward LSTM GDP forecasts based on electricity consumption for six high-income European countries. Predictions were made using a 20-quarter rolling window. In most cases, the model fails to track actual GDP values accurately, reflecting the limited short-term predictive power of electricity consumption alone

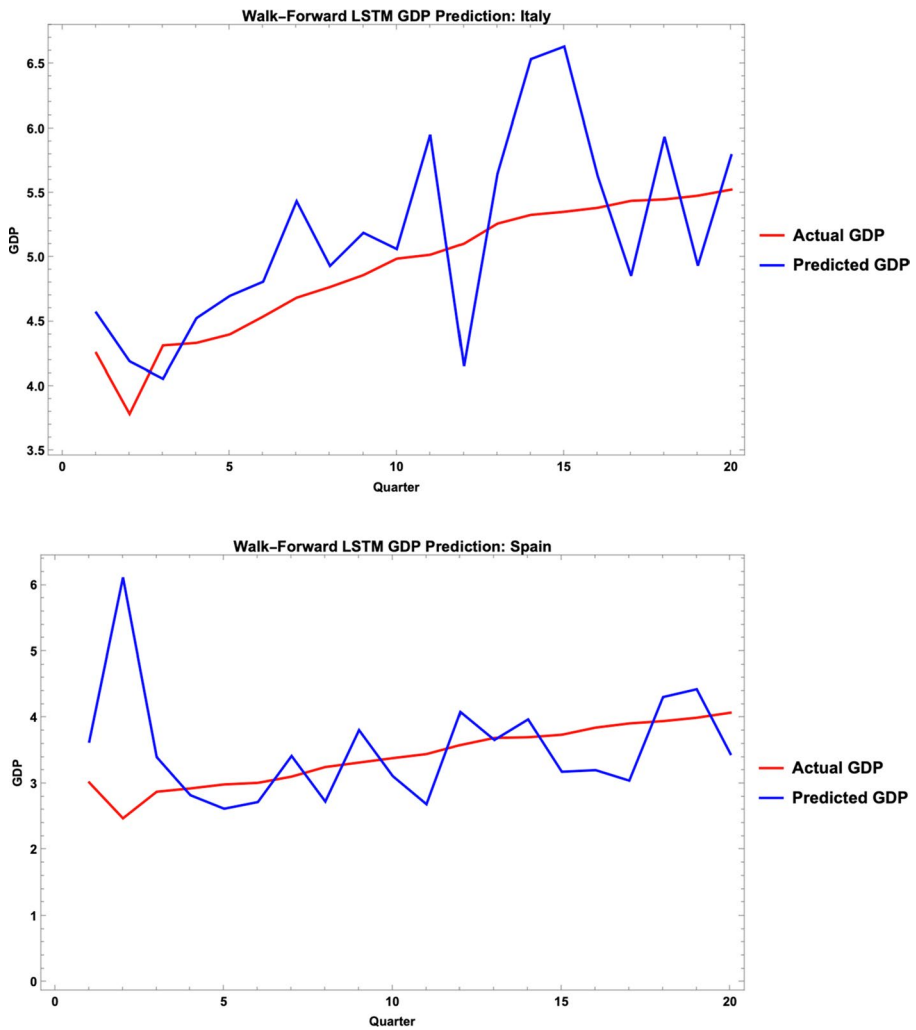


Fig. 10 (continued)

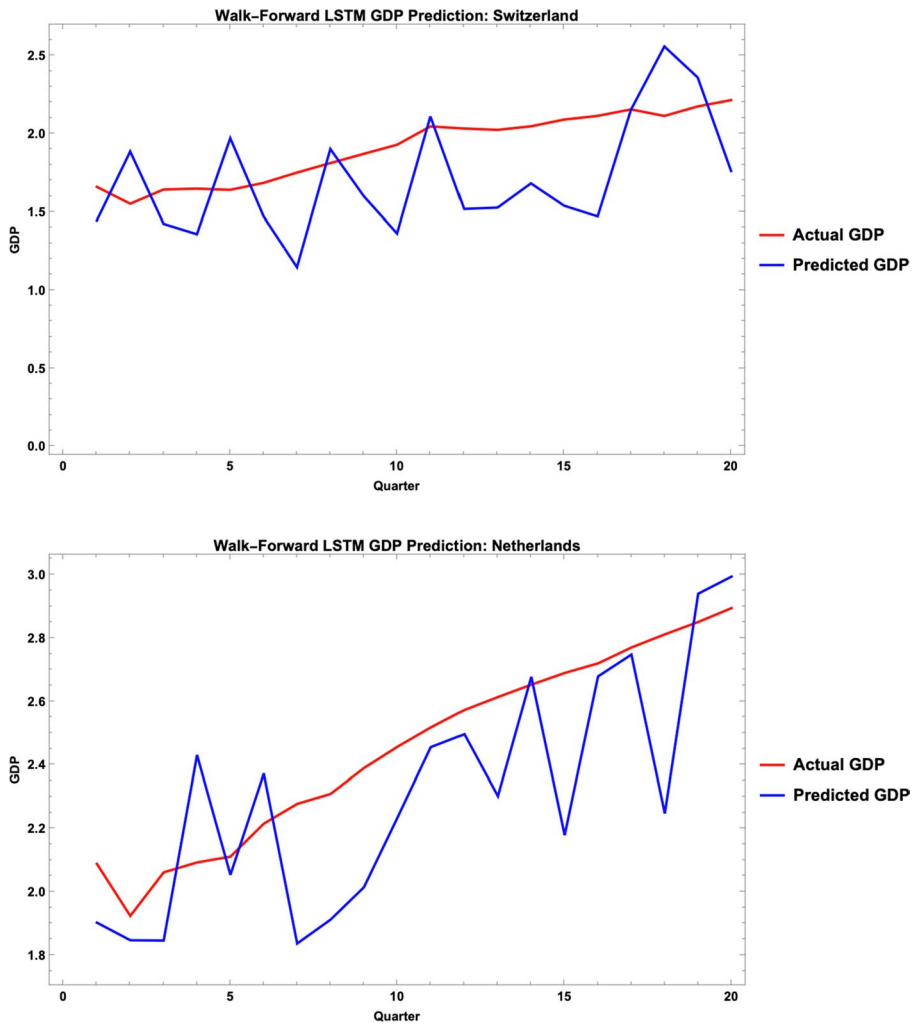


Fig. 10 (continued)

Acknowledgements The authors would like to thank the Institute of Technology and Business in České Budějovice for providing the research environment and academic support that made this study possible. We also acknowledge the use of open-access resources and tools that contributed to the analysis and modeling of this research.

Author Contributions All authors contributed to the study's conception and design. Material preparation, data collection, and initial analysis were performed by Robin Kunju Mol Raj. Marek Vochozka provided methodological guidance, critical revisions, and supervision. Yelyzaveta Apanovych assisted in data interpretation, manuscript review, and final formatting. The first draft of the manuscript was written by Robin Kunju Mol Raj, and all authors reviewed and approved the final version of the manuscript.

Funding Institute of Technology and Business in České Budějovice.

Data Availability The dataset supporting the findings of this study is publicly available in Zenodo at: <https://doi.org/10.5281/zenodo.15119740>.

Declarations

Compliance with Ethical Standards This article does not contain any studies with human participants or animals performed by any of the authors.

Consent for Publication The authors have agreed to submit and have approved the manuscript for submission.

Competing Interests The authors declare that they have no relevant financial or non-financial interests to disclose.

Clinical Trial Number Not Applicable.

Use of Artificial Intelligence Tools ChatGPT (OpenAI) was used solely to improve the language, structure, and readability of the manuscript. All intellectual content, data analysis, and conclusions were developed by the authors. The authors take full responsibility for the content of this work. For language clarity and polishing, Grammarly was used.

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References

- Abbasi, K. R., Abbas, J., & Tufail, M. (2021). Revisiting electricity consumption, price, and real GDP: A modified sectoral level analysis from Pakistan. *Energy Policy*, 149, 112087. <https://doi.org/10.1016/j.enpol.2020.112087>
- Alsaedi, Y. H., & Tularam, G. A. (2020). The relationship between electricity consumption, peak load and GDP in Saudi Arabia: A VAR analysis. *Mathematics and Computers in Simulation*, 175, 164–178. <https://doi.org/10.1016/j.matcom.2019.06.012>

- Arcabi, V., Gelo, T., Sonora, R. J., & Simurina, J. (2021). Cointegration of electricity consumption and GDP in the presence of smooth structural changes. *Energy Economics*, 97, 105196. <https://doi.org/10.1016/j.eneco.2021.105196>
- Arshad, S., & Beyer, R. C. M. (2023). Tracking economic fluctuations with electricity consumption in Bangladesh. *Energy Economics*, 123, 106740. <https://doi.org/10.1016/j.eneco.2023.106740>
- Azam, A., Rafiq, M., Shafique, M., Zhang, H., Ateeq, M., & Yuan, J. (2021). Analyzing the relationship between economic growth and electricity consumption from renewable and non-renewable sources: Fresh evidence from newly industrialized countries. *Sustainable Energy Technologies and Assessments*, 44, 100991. <https://doi.org/10.1016/j.seta.2021.100991>
- Băra, A., Oprea, S. V., & Tudorică, B. G. (2024). From the East-European regional day-ahead markets to a global electricity market. *Computational Economics*, 63(6), 2525–2557. <https://doi.org/10.1007/s10614-023-10416-0>
- Belaïd, F., & Zrelli, M. H. (2019). Renewable and non-renewable electricity consumption, environmental degradation and economic development: Evidence from Mediterranean countries. *Energy Policy*, 133, Article 110929. <https://doi.org/10.1016/j.enpol.2019.110929>
- Benkraiem, R., Lahiani, A., Miloudi, A., & Shahbaz, M. (2019). The asymmetric role of shadow economy in the energy-growth nexus in Bolivia. *Energy Policy*, 125, 405–417. <https://doi.org/10.1016/j.enpo.2018.10.060>
- Cabello Eras, J. J., Fandino, M., Gutierrez, J. M. S., Rueda Bayona, A., & German, S. (2022). The inequality of electricity consumption in Colombia. Projections and implications. *Energy*, 249, 123711. <https://doi.org/10.1016/j.energy.2022.123711>
- Caraiani, C., Lungu, C. I., Dascalu, C., & Balea, E. C. (2014). The Causality Relationship Between Energy Consumption and Gdp: The Case of Emerging European Countries. *Proceedings of the 9th International Conference Accounting and Management Information Systems (AMIS 2014)*, 620–640.
- Chandran, V. G. R., Sharma, S., & Madhavan, K. (2010). Electricity consumption-growth nexus: The case of Malaysia. *Energy Policy*, 38(1), 606–612. <https://doi.org/10.1016/j.enpol.2009.10.013>
- Colak, A. B., & Koy, A. (2023). The role of technical indicators in the intraday prediction of stock markets: Artificial neural network models for Borsa Istanbul. *Scientia Iranica*. <https://doi.org/10.24200/sci.2023.58490.5752>
- Cui, W., Li, J., Xu, W., & Gunalp, B. (2021). Industrial electricity consumption and economic growth: A spatio-temporal analysis across prefecture-level cities in China from 1999 to 2014. *Energy*, 222, Article 119932. <https://doi.org/10.1016/j.energy.2021.119932>
- Cui, C. R., Kim, J. T., Kim, S., Lee, W. J., & Sunwoo, H. Y. (2022). Green commitment and electricity consumption. *Economics Letters*, 214, 110452. <https://doi.org/10.1016/j.econlet.2022.110452>
- Dey, S. R., & Tareque, M. (2020). Electricity consumption and GDP nexus in bangladesh: A time series investigation. *Journal of Asian Business and Economic Studies*, 27(1), 35–48. <https://doi.org/10.1108/JABES-04-2019-0029>
- Esmaceli, P., & Rafei, M. (2021). Dynamics analysis of factors affecting electricity consumption fluctuations based on economic conditions: Application of SVAR and TVP-VAR models. *Energy*, 226, 120340. <https://doi.org/10.1016/j.energy.2021.120340>
- Georgescu, I. A., Oprea, S. V., & Băra, A. (2024). Investigating the relationship between macroeconomic indicators, renewables and pollution across diverse regions in the globalization era. *Applied Energy*, 363, 123077. <https://doi.org/10.1016/j.apenergy.2024.123077>
- Ghosh, S. (2009). Electricity supply, employment and real GDP in India: Evidence from cointegration and granger-causality tests. *Energy Policy*, 37(8), 2926–2929. <https://doi.org/10.1016/j.enpol.2009.03.022>
- Guefano, S., Bozorg, M., Carpita, M., Etet-Baha, P. D., Koffi, F. L. D., Tamba, J. G., & Etame, J. (2023). Causality between residential electricity consumption and explanatory factors. *Energy Strategy Reviews*, 49, 101155. <https://doi.org/10.1016/j.esr.2023.101155>
- Han, H., & Radwan, A. (2021). Economic and social structure and electricity consumption in Egypt. *Energy*, 231, 120962. <https://doi.org/10.1016/j.energy.2021.120962>
- Hao, Y., Zhang, T., Jing, L., & Xiao, L. (2019). Would the decoupling of electricity occur along with economic growth? Empirical evidence from the panel data analysis for 100 Chinese cities. *Energy*, 180, 615–625. <https://doi.org/10.1016/j.energy.2019.05.014>
- Hdom, H. A. D., & Fuinhas, J. A. (2020). Energy production and trade openness: Assessing economic growth, CO2 emissions and the applicability of the cointegration analysis. *Energy Strategy Reviews*, 30, 100488. <https://doi.org/10.1016/j.esr.2020.100488>

- Ikegami, M., & Wang, Z. (2016). The long-run causal relationship between electricity consumption and real GDP: Evidence from Japan and Germany. *Journal of Policy Modeling*, 38(5), 767–784. <https://doi.org/10.1016/j.jpolmod.2016.10.007>
- Kolin, S. K., Sedlar, D. K., & Kurevija, T. (2021). Relationship between electricity and economic growth for long-term periods: New possibilities for energy prediction. *Energy*, 228, 120539. <https://doi.org/10.1016/j.energy.2021.120539>
- Korkmaz, O. (2022). What is the role of the rents in energy connection with economic growth for China and the United States?? *Resources Policy*, 75, 102517. <https://doi.org/10.1016/j.resourpol.2021.102517>
- Koy, A., & Çolak, A. B. (2023). Predicting stock market index and credit default swap spreads using artificial intelligence and determining nonlinear relations. *Archives of Advanced Engineering Science*, 1–12. <https://doi.org/10.47852/bonviewAAES32021366>
- Koy, A., & Çolak, A. B. (2024). The Intraday High-Frequency Trading with Different Data Ranges: A Comparative Study with Artificial Neural Network and Vector Autoregressive Models. *Archives of Advanced Engineering Science*, 2(3), 3. <https://doi.org/10.47852/bonviewAAES32021325>
- Kraft, J., & Kraft, A. (1978). On the relationship between energy and GNP. *The Journal of Energy and Development*, 3(2), 401–403.
- Krulický, T., Jura, A., Kunju, M., & Raj, R. (2024). Navigating the dynamics of Brent crude oil prices: Factors, trends, and insights. *Acta Montanistica Slovaca*, 29(v29/i1), 86–102. <https://doi.org/10.46544/AMS.v29i1.08>
- Kunju, M., & Raj, R. (2025). *Electricity consumption and gdp of top economic country in europe*. <https://doi.org/10.5281/zenodo.15119741>
- Laborda, J., Ruano, S., & Zamanillo, I. (2023). Multi-country and multi-horizon GDP forecasting using temporal fusion transformers. *Mathematics*, 11, 2625. <https://doi.org/10.3390/math11122625>
- Liddle, B., Parker, S., & Hasanov, F. (2023). Why has the OECD long-run GDP elasticity of economy-wide electricity demand declined? Because the electrification of energy services has saturated. *Energy Economics*, 125, 106832. <https://doi.org/10.1016/j.eneco.2023.106832>
- Lin, B., & Raza, M. Y. (2021). Analysis of electricity consumption in Pakistan using index decomposition and decoupling approach. *Energy*, 214, 118888. <https://doi.org/10.1016/j.energy.2020.118888>
- Liu, Y., Wang, L., & Deng, Y. (2024). Electricity generation and economic performance: On the 2021 Texas power crisis. *Electricity Journal*, 37(1), Article 107367. <https://doi.org/10.1016/j.tej.2024.107367>
- Massa, R., & Rosellon, J. (2020). Linear and nonlinear Granger causality between electricity production and economic performance in Mexico. *Energy Policy*, 142, Article 111476. <https://doi.org/10.1016/j.enpol.2020.111476>
- Narayan, P. K., Narayan, S., & Popp, S. (2010). Does electricity consumption panel Granger cause GDP? A new global evidence. *Applied Energy*, 87(10), 3294–3298. <https://doi.org/10.1016/j.apenergy.2010.03.021>
- Navarro, C. E. B., Alvarez-Quiroz, V. J., Sampi, J., & Sanchez, A. A. A. (2023). Does economic growth promote electric power consumption? Implications for electricity conservation, expansive, and security policies. *Electricity Journal*, 36(1), Article 107235. <https://doi.org/10.1016/j.tej.2023.107235>
- Ohler, A., & Feters, I. (2014). The causal relationship between renewable electricity generation and GDP growth: A study of energy sources. *Energy Economics*, 43, 125–139. <https://doi.org/10.1016/j.eneco.2014.02.009>
- Oprea, S. V., Bâra, A., & Georgescu, I. (2025). An economic and environmental perspective of the economic growth, emissions, foreign investment and renewables integration in the Black Sea region. *Kybernetes, ahead-of-print(ahead-of-print)*. <https://doi.org/10.1108/K-09-2024-2451>
- Ouedraogo, I. M. (2010). Electricity consumption and economic growth in Burkina Faso: A cointegration analysis. *Energy Economics*, 32(3), 524–531. <https://doi.org/10.1016/j.eneco.2009.08.011>
- Ouedraogo, N. S. (2013). Energy consumption and economic growth: Evidence from the economic community of West African States (ECOWAS). *Energy Economics*, 36, 637–647. <https://doi.org/10.1016/j.eneco.2012.11.011>
- Pandellara, D., Kristjanpoller, W., Michell, K., & Minutolo, M. C. (2022). A fuzzy regression causality approach to analyze relationship between electrical consumption and GDP. *Energy*, 239, 122459. <https://doi.org/10.1016/j.energy.2021.122459>
- Rahman, Z. U., Khattak, S. I., Ahmad, M., & Khan, A. (2020). A disaggregated-level analysis of the relationship among energy production, energy consumption and economic growth: Evidence from China. *Energy*, 194, 116836. <https://doi.org/10.1016/j.energy.2019.116836>

- Sadorsky, P. (2012). Information communication technology and electricity consumption in emerging economies. *Energy Policy*, 48, 130–136. <https://doi.org/10.1016/j.enpol.2012.04.064>
- Saldivia, M., Kristjanpoller, W., & Olson, J. E. (2020). Energy consumption and GDP revisited: A new panel data approach with wavelet decomposition. *Applied Energy*, 272, 115207. <https://doi.org/10.1016/j.apenergy.2020.115207>
- Tran, B. L., Chen, C. C., & Tseng, W. C. (2022). Causality between energy consumption and economic growth in the presence of GDP threshold effect: Evidence from OECD countries. *Energy*, 251, 123902. <https://doi.org/10.1016/j.energy.2022.123902>
- Wu, C. F., Wang, C. M., Chang, T., & Yuan, C. C. (2019). The nexus of electricity and economic growth in major economies: The United States-India-China triangle. *Energy*, 188, 116006. <https://doi.org/10.1016/j.energy.2019.116006>
- Wu, W., Kanamori, Y., Zhang, R., Zhou, Q., Takahashi, K., & Masui, T. (2021). Implications of declining household economies of scale on electricity consumption and sustainability in China. *Ecological Economics*, 184, 106981. <https://doi.org/10.1016/j.ecolecon.2021.106981>
- Yang, Y. (2010). Electricity Consumption and Economy Growth: Cointegration Technique Applied in China. *2010 Asia-Pacific Power and Energy Engineering Conference (APPEEC)*. Asia-Pacific Power and Energy Engineering Conference (APPEEC), New York. <https://www.webofscience.com/wos/woscc/full-record/WOS:000396400801035>
- Zhang, M., Zhang, K., Hu, W., Zhu, B., Wang, P., & Wei, Y. M. (2020). Exploring the climatic impacts on residential electricity consumption in Jiangsu, China. *Energy Policy*, 140, 111398. <https://doi.org/10.1016/j.enpol.2020.111398>
- Zhang, G., Nuruzzaman, M., & Su, B. (2021). Nexus between household energy consumption and economic growth in Bangladesh (1975–2018). *Energy Policy*, 156, 112420. <https://doi.org/10.1016/j.enpol.2021.112420>

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Authors and Affiliations

Robin Kunju Mol Raj¹  · Marek Vochozka¹  · Yelyzaveta Apanovych¹ 

✉ Robin Kunju Mol Raj
32110@mail.vstecb.cz

Marek Vochozka
vochozka@mail.vstecb.cz

Yelyzaveta Apanovych
apanovych@mail.vstecb.cz

¹ Institute of Technology and Business in České Budějovice, School of Expertness and Valuation, Okružní 517/10, České Budějovice 37001, Czech Republic